

Study 2 Writeup

Feature generation key takeaways

We use two approaches for generating features:

1. **BERTopic:** Clusters post embeddings and generates topics based on the substance of the text.
2. **LLM-driven feature generation:** generates features based on both the substance of text and the civility of conversation.

The key takeaway across both these methods is that:

- The topic, substance, or political lean of a post are weak predictors of keep/remove behavior
- The civility of conversation (i.e., the toxicity of language) is a strong predictor.

BERTopic on the original posts

We focus the BERTopic analysis on the original posts. The mirroring operation did change some of the topics from the original post. We try to keep the same stance and political intensity, but there are different topics that are triggers for left-leaning and right-leaning groups. We add a follow-up set of analyses later on reviewing how the mirroring approach changed the topics. We also use all keep/remove decisions per post, rather than taking a majority vote. For convention's sake, we measure the overall keep rate, rather than the remove rate.

We see more well-defined BERTopic results from the larger combined Study 2 dataset than we did in the first run given our larger dataset and the increased number of labels per post.

Overview

Below is a 2-D visualization of the posts clustered by their key topics. An interactive visualization can be found at [this link](#).

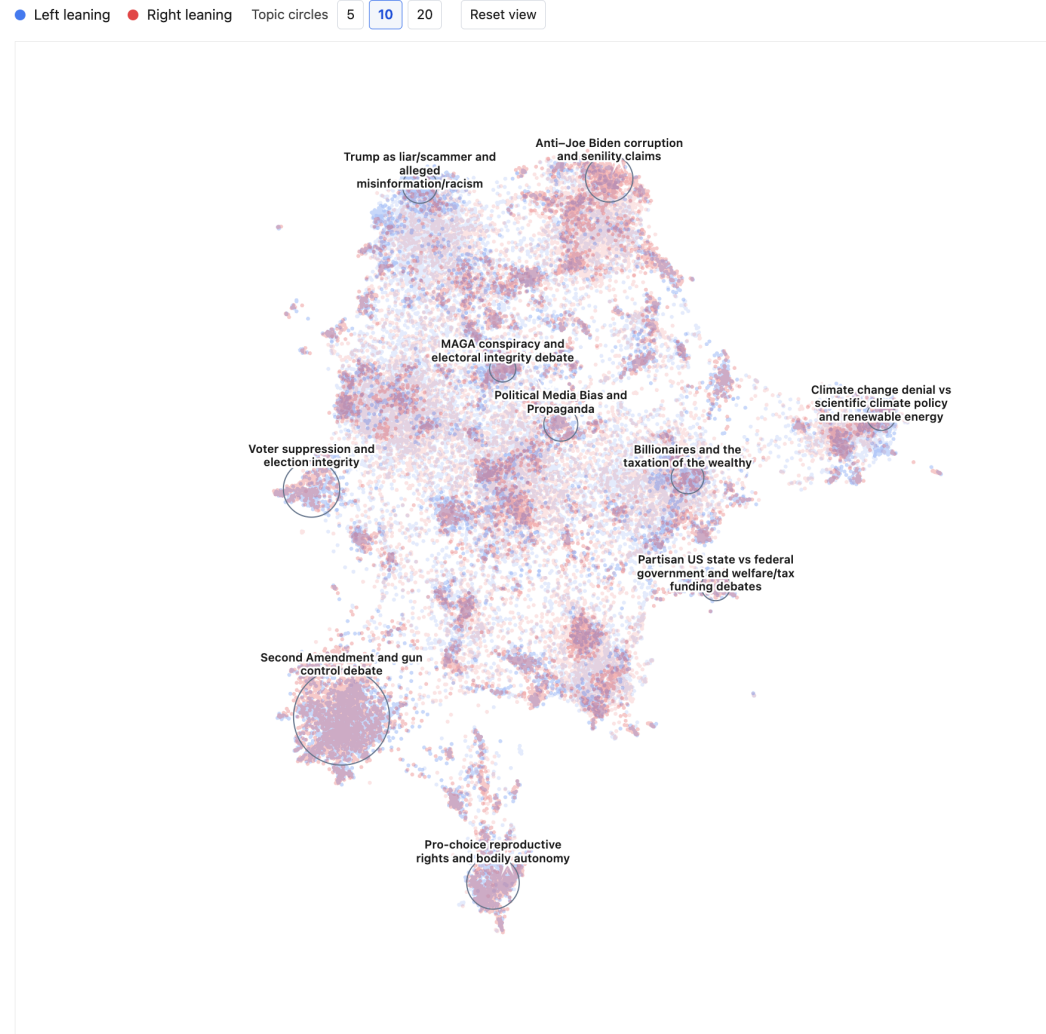


Figure 1: BERTopic Cluster Map

Which topics most commonly appear?

Topic	Posts	Keep rate
Climate and fossil fuels	1,122	81.8%
Border and immigration	862	75.8%
MAGA and institutions	851	59.1%
Abortion access	690	75.2%
Trump media and lying	636	59.3%
Criticism of Democrats	598	69.3%

Topic	Posts	Keep rate
Criticism of Republicans	566	59.0%
Biden and Trump blame	358	67.6%
Billionaires and taxes	326	76.6%
Abolish ICE	259	62.1%

Which topics are more common in originally left leaning posts, and which in right leaning posts?

Each post and its mirror are shown, so users see both the left-leaning and right-leaning versions. We analyze, based on the original posts, which topics tend to be more common in left-leaning vs. right-leaning posts.

Some topics appear much more often in left- or right-leaning posts

Share of posts within each political-leaning group, by topic

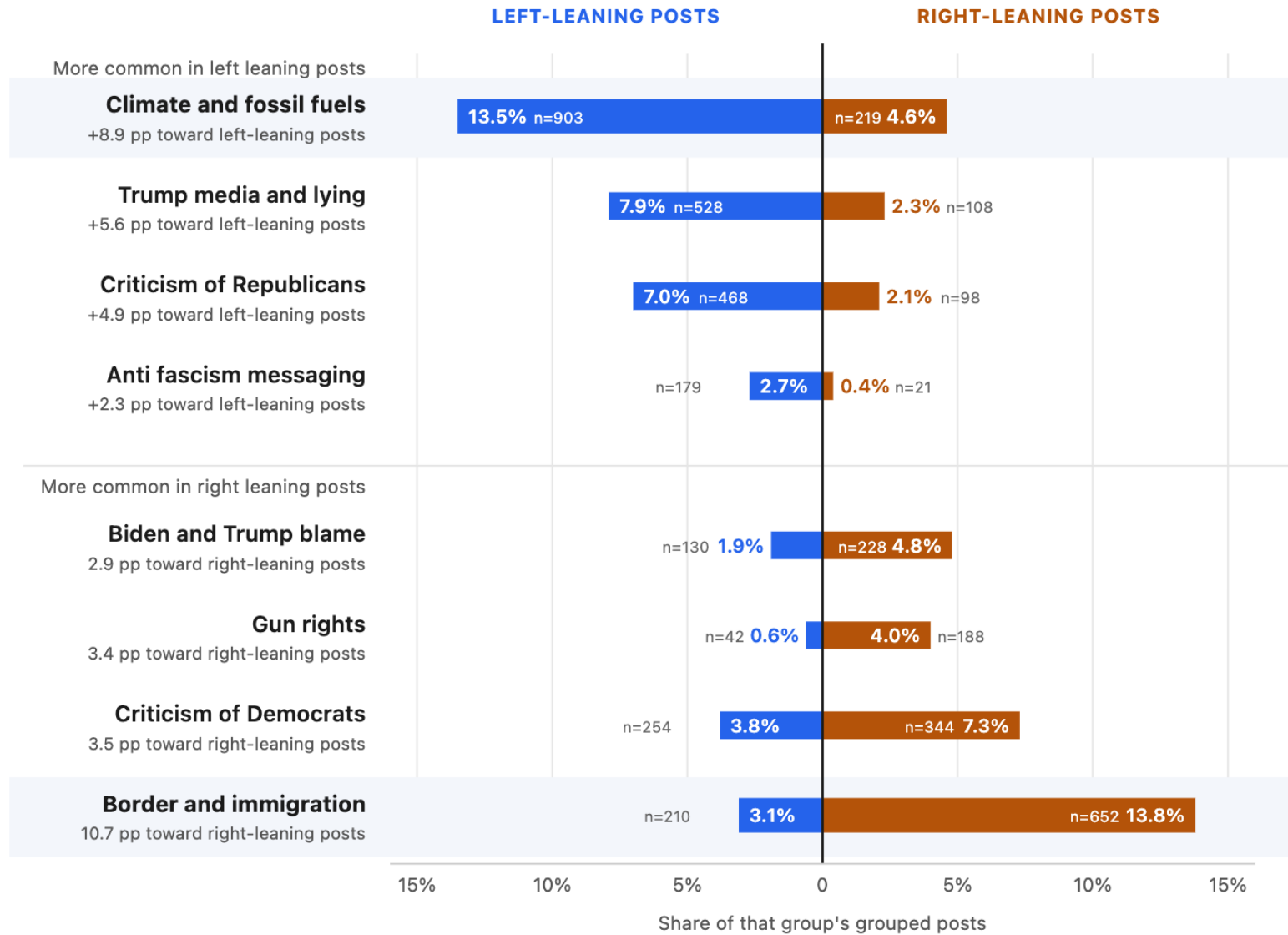


Figure 2: Topics by Left/Right-Leaning

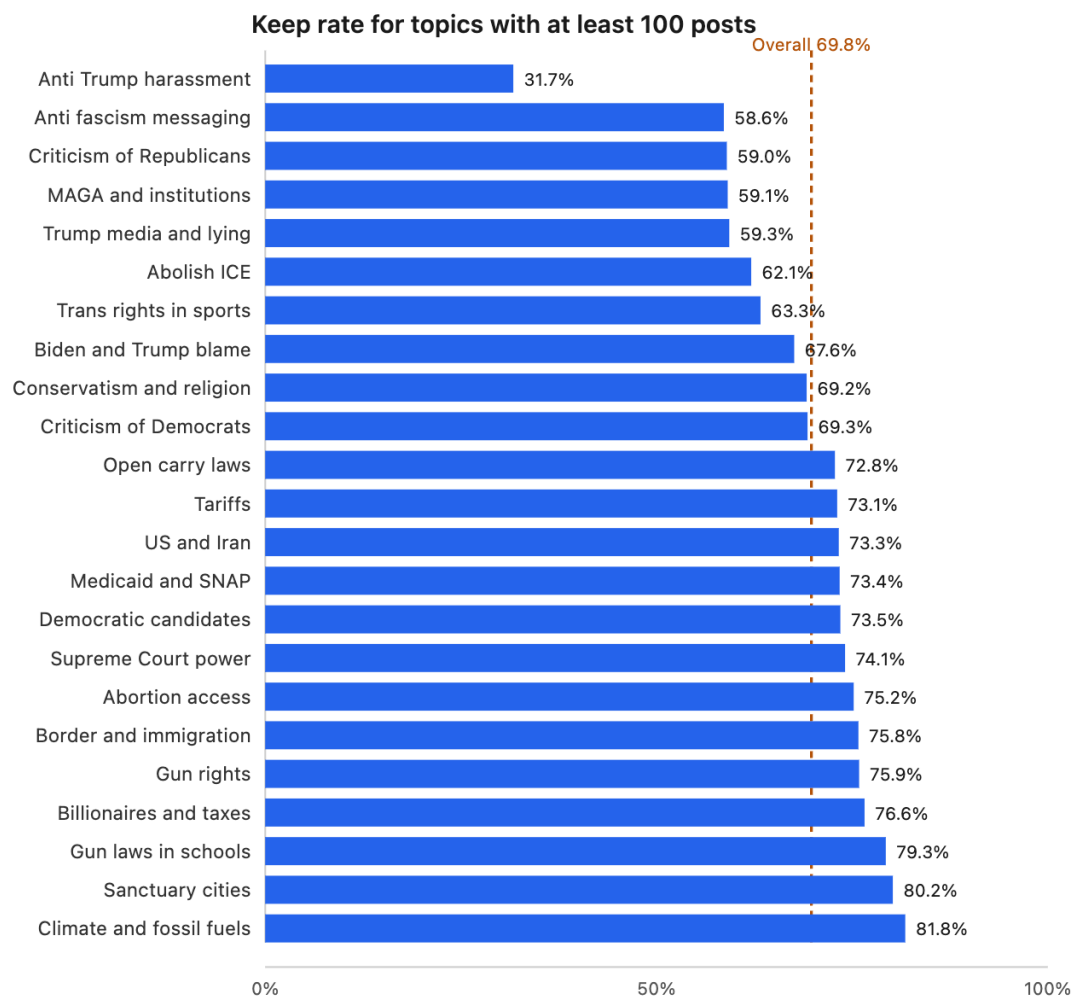
Which topics are more common at low, medium, and high toxicity?

Low toxicity posts are more often about climate, abortion, and billionaires and taxes. Climate alone is 20.5% of low toxicity grouped posts and 1.6% of high toxicity grouped posts. High toxicity posts are more often attacks on Trump, MAGA and institutions, and criticism of Republicans. Criticism of Trump’s media behavior is 13.9% of high toxicity grouped posts and 1.1% of low toxicity grouped posts. Anti Trump harassment is almost entirely in the high toxicity sample (140 of 144 posts).

Topic	Low	Medium	High
Climate and fossil fuels	20.5% (604)	8.5% (471)	1.6% (47)
Abortion access	8.2% (240)	7.0% (389)	2.1% (61)
Billionaires and taxes	4.8% (141)	2.1% (117)	2.4% (68)
Trump media and lying	1.1% (33)	3.6% (200)	13.9% (403)
Anti Trump harassment	0% (0)	0.1% (4)	4.8% (140)
MAGA and institutions	3.3% (98)	7.9% (439)	10.9% (314)
Criticism of Republicans	2.2% (66)	4.9% (272)	7.9% (228)

Do some topics get removed more often?

Among the 23 topics with at least 100 posts, the keep rate runs from 31.7% to 81.8%. The topics most commonly removed include anti-Trump harassment and anti-fascism messaging, while the topics most commonly kept include climate change and sanctuary cities.



Each bar is one topic with at least 100 posts. The dashed line is the overall keep rate, 69.8%. The percentage is the average share of raters who voted keep.

Figure 3: Keep Rate by Topic

Which topics did Democratic/Republican raters keep, and which did they remove?

Democratic raters kept 69.4% of 52,864 votes. Republican raters kept 70.0% of 48,970 votes. The topics they were most likely to remove, and the topics they were most likely to keep, are almost the same list. Both groups removed anti Trump harassment most often, at about 32%. Both groups kept climate and fossil fuels most often among large topics, at about 82%.

Where the two groups differ, the gap is generally small. Across 26 topics with at least 200 votes from each group, the typical gap is under 2 points. The largest gaps are:

- Open carry laws: Democratic raters kept 75.8% of 289 votes, and Republican raters kept 68.3% of 249 votes.
- Supreme Court: Republican raters kept Supreme Court posts at 77.6% (308 votes), and Democratic raters kept them at 71.5% (330 votes).

Keep/remove behavior for Democrats Here are the topics Democratic raters were most likely to remove, among topics with at least 200 of their votes.

Topic	Keep rate	Votes
Anti Trump harassment	31.5%	444
MAGA and institutions	57.6%	2,308
Criticism of Republicans	59.1%	1,500
Trump media and lying	59.4%	1,788
Anti fascism messaging	59.7%	571

Here are the topics Democrat raters were most likely to keep:

Topic	Keep rate	Votes
Gun policy debate	82.4%	262
Climate and fossil fuels	82.2%	2,872
Gun laws in schools	79.5%	410
Sanctuary cities	78.6%	266
Gun rights	77.6%	602

Keep/remove behavior for Republicans Here are the topics Republican raters were most likely to remove, among topics with at least 200 of their votes.

Topic	Keep rate	Votes
Anti Trump harassment	31.9%	282
Anti fascism messaging	57.7%	478
Criticism of Republicans	59.1%	1,377
Trump media and lying	59.2%	1,473
MAGA and institutions	60.8%	2,059

Here are the topics Republican raters were most likely to keep:

Topic	Keep rate	Votes
Climate and fossil fuels	81.5%	2,893
Sanctuary cities	80.6%	289
Gun policy debate	80.2%	242
Gun laws in schools	79.1%	369
Mail and elections	78.2%	257

Does the stance of the post affect the keep/remove rate? The stance of the post barely moves the overall keep rate. Left leaning posts are kept at 69.0% (11,386 posts). Right leaning posts are kept at 70.9% (8,377 posts). Within topics, certain topics present in left-leaning or right-leaning posts are kept more often:

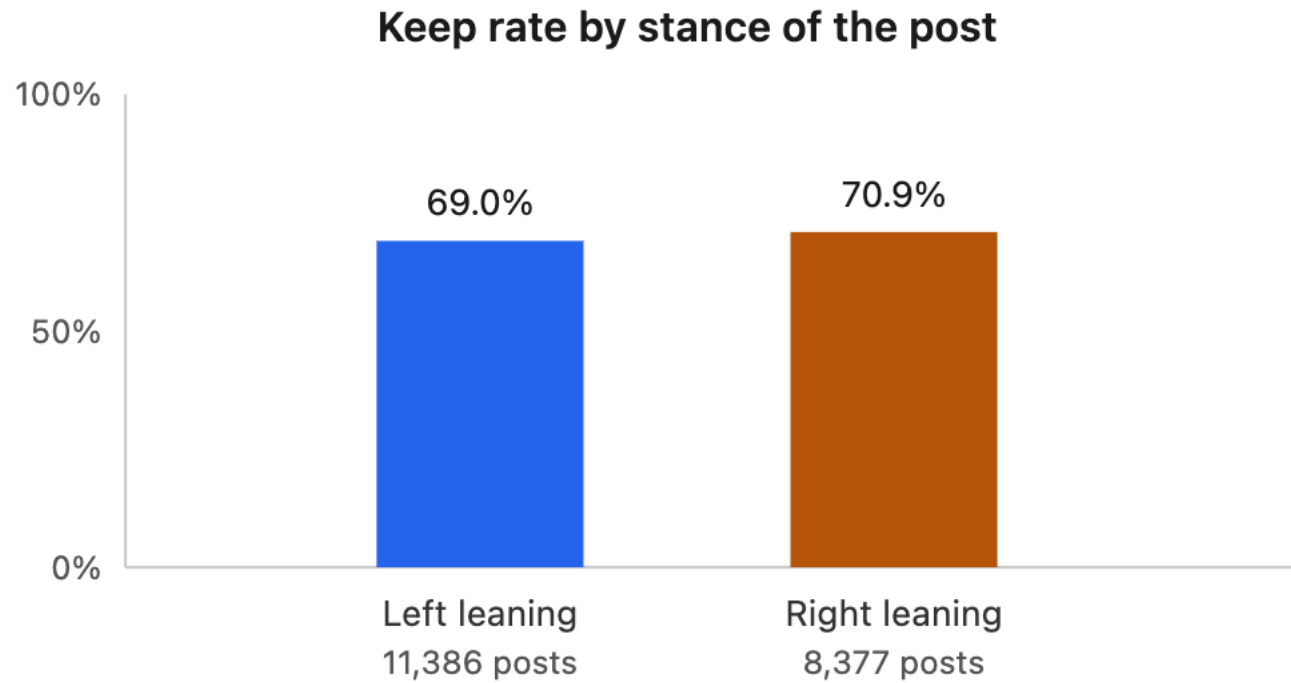


Figure 4: Keep Rate by Political Lean

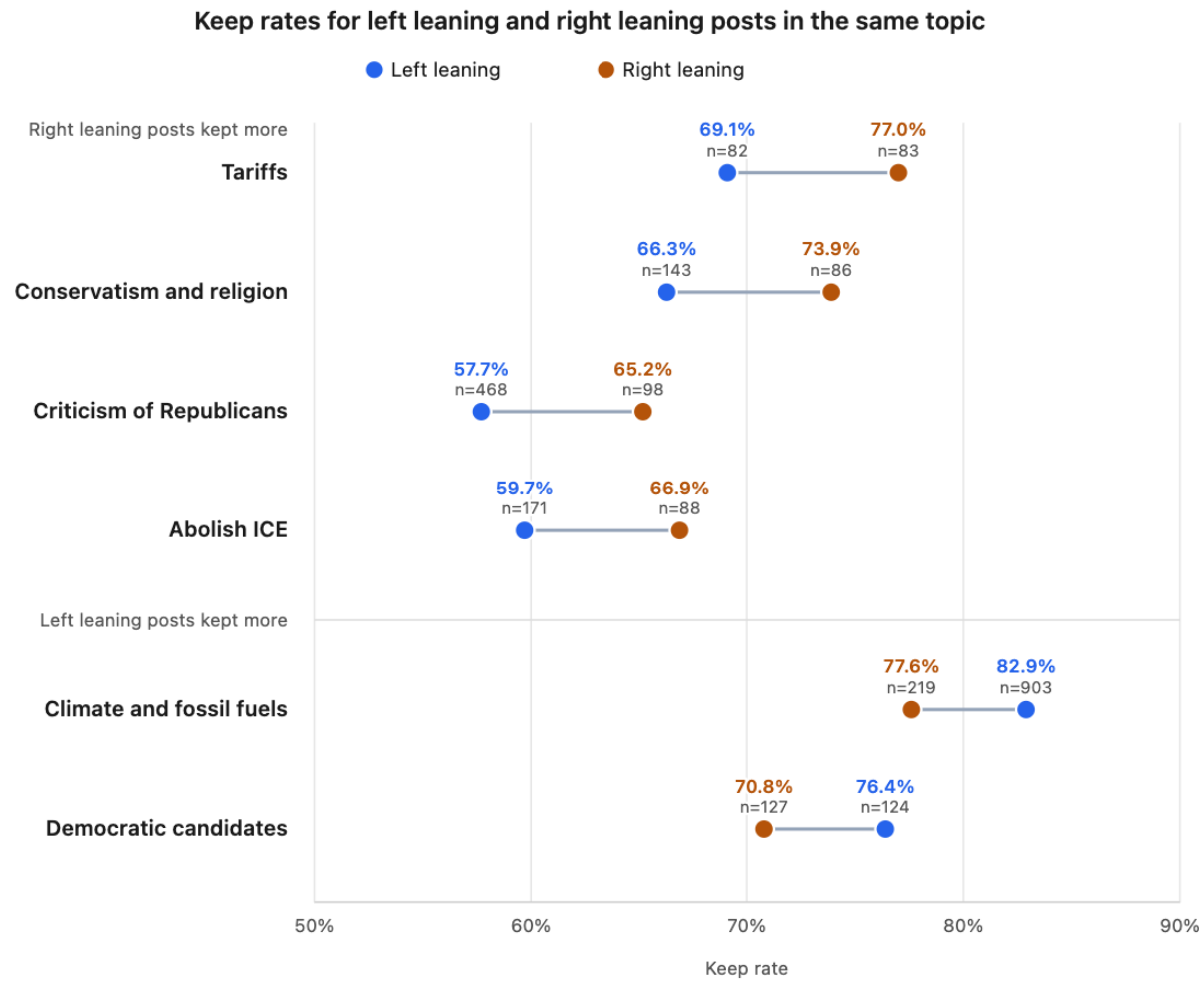


Figure 5: Keep Rate by Topic and Political Lean

Does the toxicity of the post affect the keep/remove rate? Toxicity moves the keep rate much more than stance does. Low toxicity posts are kept at 84.7% (4,892 posts). Medium toxicity posts are kept at 72.6% (9,888 posts). High toxicity posts are kept at 49.8% (4,983 posts).

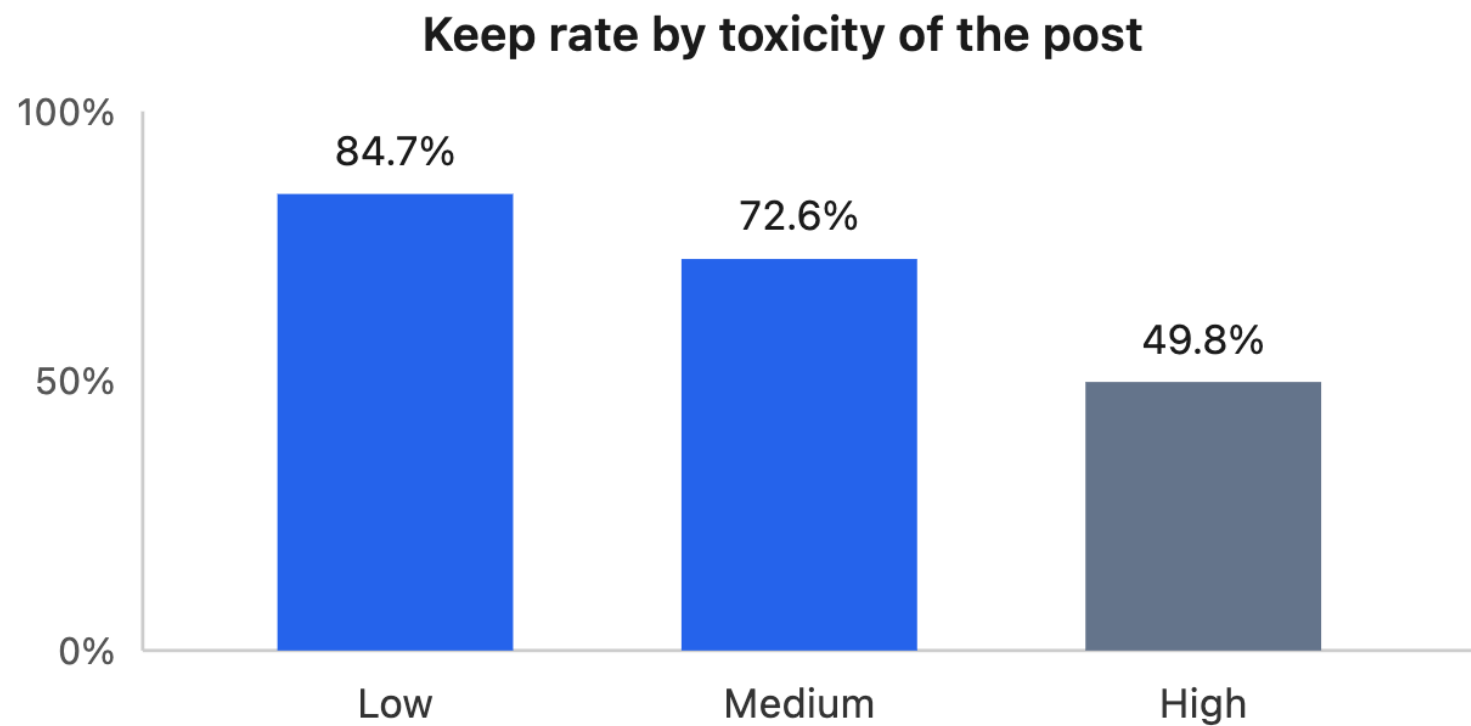


Figure 6: Keep Rate by Toxicity

This trend is generally true across topics as well:

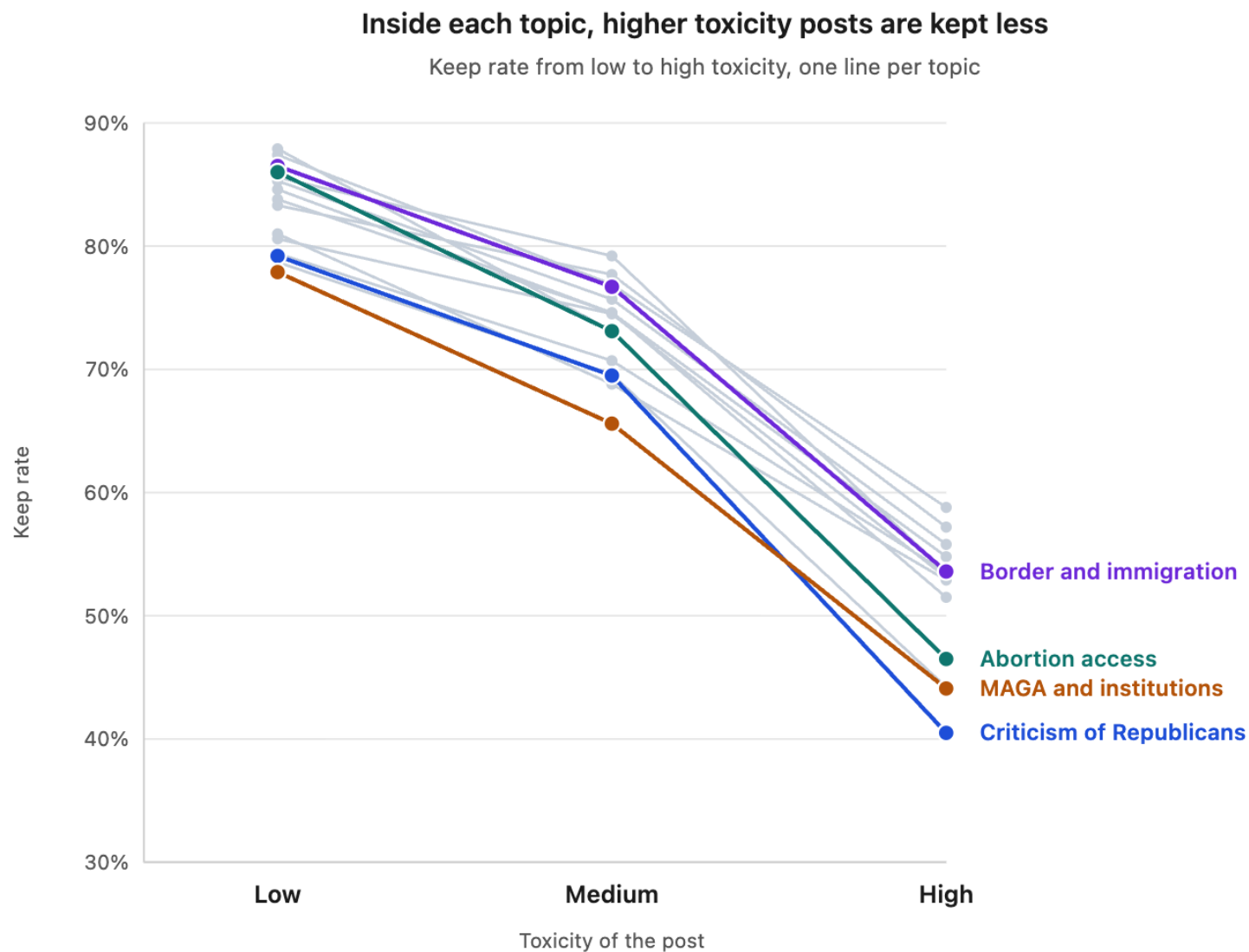


Figure 7: Keep Rate by Toxicity and Topic

LLM-based feature generation

We then use LLMs to perform feature extraction. We follow a protocol developed by Google DeepMind on how to use LLMs for automated feature extraction.

Methods

1. **Setup:** We assign a keep/remove label to each original + mirror combination based on the majority label.

2. **Use an LLM to mine features:** We pass in batches of 10 pairs of posts that were majority keep and 10 pairs of posts that were majority remove. We then ask an LLM to extract features to distinguish posts that were kept and posts that were removed. We do this across a few categories of features (see below).
3. **Embed the feature records and cluster them:** We embed the features and then use K-Means to generate clusters. We do this instead of recursively asking an LLM to generate a feature category given a list of features since we want a sense of “global similarity” across all features. We choose K-Means since HDBSCAN rendered unstable estimates.
4. **Name each cluster:** We take each cluster and pass them to an LLM to generate a human-readable label and description of each feature cluster. This generates a compiled list of feature groups.
5. **Label every post against the features:** We label each post against each feature group. This means that a post can have multiple features, which makes this approach more feature-rich and expressive than BERTopic.

Categories of features

These are the categories that we gave to the LLM when asking them to mine for features in the original posts:

1. **Surface and lexical:** This category is about how the post is written, not what claim it makes. It covers length, slang, heavy punctuation, all-caps emphasis, profanity, hashtags and account mentions, and a high density of proper names. Examples include emphatic typography, profane derogatory insults, colloquial language and insults, and hashtags and account mentions.
2. **Topic and subject matter:** This category is about the subject of the post. Examples include a policy area (guns, climate, immigration, abortion, elections), a specific event or bill, a geographic scope, a historical analogy, and culture-war salience.
3. **Semantic content:** This category is about the kind of claim the post makes. Examples include causal claims, moral language, a factual claim versus speculation, conspiracy, a claim that a group is being persecuted, a policy prescription, and a cost-benefit argument.
4. **Pragmatics and communicative intent:** This category is about what the post is doing to the reader. Examples include sarcasm, mockery, a call to action, persuasion, venting, hedging, and outrage.
5. **Target and directionality:** This category is about who the post attacks or praises, and which political side it points at. Examples include the type of actor criticized or praised, a left/right cue, us-versus-them framing, and elite-versus-populist framing.
6. **Compositional and syntactic structure:** This category is about the shape of the sentences. Examples include if-then conditionals, contrast with “but” or “however,” rhetorical questions, parallel repetition, lists, quoted or attributed speech, and direct address in the second person.

Discovered categories

After mining features, embedding, and naming each resulting feature cluster, here are the feature categories that were discovered:

Name	Definition	Examples
Blanket Demonization of Political Opponents	Sweeping, unqualified portrayals of an opposing political group as inherently stupid, immoral, corrupt, dangerous, subhuman, or traitorous.	1. All maga people are butthurt morons 2. Trumpers are the dumbest humans to have ever lived, they have no idea about anything. 3. Wear the downvotes as a badge of honor. Leftists are fucking losers and nothing they say or do matters to any sane human. 4. Another team called “The MAGAs”. A bunch of toothless, sister-fucking idiots as mascots.

Name	Definition	Examples
Brief Slogan-Like Insult Attacks	Pairs feature very short, blunt, insult-led statements or fragments that make a denunciatory claim with little or no supporting elaboration.	1. That's because Trump voters are idiots. 2. Because trump supporters are bitter garbage 3. Hypocrisy is the right's only "principle". 4. trump - "Who would hire these losers"
Broad Partisan Out-Group Targeting	The pair targets or refers to ordinary members, supporters, voters, or whole populations defined by broad partisan or ideological identities rather than only specific leaders, institutions, or policies.	1. All liberal white women support mass illegal immigration and they all demand diversity equity and inclusion. 2. Yep, fuck em. I'd still like to see progressive change, but my empathy no longer extends to the dorks that voted for their own nightmare. 3. > Most of the maga males probably think it's awesome and support that shit. They absolutely do. 4. Conservative Christians r simply self righteous delusional psychopaths who believe they have divine permission to lie cheat+steal bec it's all for their deity ...
Culture-War Policy Issues	Posts focus on contested partisan policy debates over guns, immigration and border enforcement, abortion and reproductive rights, policing, or related rights-based social issues.	1. Republicans were right! Our second amendment MUST be protected!!! 2. Pro-life, and it's probably my single most important political issue. 3. Why do you think the Trump administration is not doing something about this? We need immigration law reform ALONG WITH deportation. 4. Also stop trying to abolish or defund or get rid of things you don't like. Most people don't like extreme measures. Defund the police backfired and I can see the Abolish ICE rhetoric backfiring at some point in the future
Culture-War Security and Out-Group Threats	Posts frame immigration, policing, guns, identity issues, religion, or political unrest as existential threats to public safety, national identity, or social order.	1. Those Muslims are infiltrating developed nations in order to invade them. Over 85% of the world's refugees are Muslims, mostly adult men. ... 2. Russians are the world's worst illegal immigrants: They invade with missiles, drones, and tanks. And against this tide of violent illegal immigration, Ukrainians say, "you shall not pass." Ukrainians are protecting the West. 3. ... America will not just roll over. We will not allow our country to become a Muslim country. ... 4. It's not the first stages. We're well into Trump's fascist police state. Resist, and don't stop resisting.

Name	Definition	Examples
Dense Profanity and Vulgar Insults	The pair contains frequent or repeated explicit profanity, obscenity, vulgarity, or derogatory insult terms directed at people or groups.	1. Fuck Texas. Fuck Idaho. Fuck MAGA. Fuck Trump. Fuck Cruz. 2. Fuck off. Fuck every possible way off. Fuck off to the fertile fuck-fields of Fuckoffizona. Fuck the Republicans, and fuck you for collaborating with those traitors ... 3. I like him even more now. Fuck Republicans, fuck Republican voters, fuck billionaires, fuck anyone even sniffing conservatism/MAGA bullshit. 4. Fuck ICE. Fuck Trump. Fuck this corrupt ass DoJ. Fuck two-thirds of the Supreme Court. And fuck anyone else enabling this shit.
Derogatory Labels and Epithets	The post pair uses insulting, demeaning, slur-like, or dehumanizing labels to characterize people or political groups.	1. How about the uneducated low-skill redneck welfare garbage Alabamans mooching off the wealth created by the ‘coastal elites’ 2. The r-tard right is, unsurprisingly, infested with unironic neo nazis. The silver lining is that these people are often not even Americans so it balances out. 3. I blame Christian nationalism for this. American Christian nationalists (I call them the Y’all Qaeda) would rather turn America into a theocratic hellhole than actually love thy neighbors, as Jesus stated to do. 4. ... Republican alpha culture literally makes every Republican but Trump an alpha. There’s only one alpha in alpha society. The rest are beta cucks.
Direct Second-Person Hostility	Posts directly address an individual or group as “you,” “you guys,” or similar terms while using insults, taunts, accusations, threats, or hostile commands.	1. Fuck you Trump from a retired disabled combat vet! 2. i hope i never have a child as pathetic and asinine as you. 3. @TomSteyer You are an absolute disgrace and you are putting women’s rights back to the stone age. 4. If you are an ICE agent fuck you. I hope every moment of your life is suffering. ...
Electoral Strategy and Political Consequences	Posts that predict, warn about, or explain how parties, politicians, voters, policies, or messaging will affect electoral outcomes, public support, political behavior, or longer-term political consequences.	1. Basically. Trump probably would’ve won in 2020 without the pandemic dropping living standards. That’s Dems only chance 2. Trump is uniting the country against him. Midterms are going to be an absolute massacre for republicans. 3. So what. Republicans will win again in 28 if a good candidate for the left cannot be found. 4. People forget that evangelicals make a non-insubstantial part of the Republican base, so being anti-Israel is kissing your political career goodbye.

Name	Definition	Examples
Elite-versus-Public Political Framing	Frames political or economic conflict as powerful elites, institutions, or wealthy interests opposing ordinary people, workers, voters, or taxpayers.	1. There was never a better time to stay the course. This man is the only one who cares about the people not the ruling class. 2. ... you didn't need Karl Marx to come around to tell you that the rich organized themselves against the workers. ... 3. Good luck with that. Congress ain't gonna let it happen. They don't work for you. They work for their rich friends. 4. Of course Senators and Congressfolk still get their Cadillac tax payer funded healthcare.
Escalatory Partisan Hostility	Posts use inflammatory us-versus-them rhetoric, contempt, threats, catastrophe claims, or gloating to intensify hostility toward political opponents rather than argue policy.	1. if Democrats ever get control again they need to play as dirty as Trump. Let the red states starve. 2. Trump supporters still: So glad to have trump as our president. Democrats would have led us into ww3 by now 3. Probably because they know that Trump will order their deaths once he finishes his dictatorial power grab. It's the same reason 2A liberals haven't started using the 2nd amendment for its intended purpose: they don't want to die. 4. Calling on all Democrats to arm themselves, like 10 years ago when it was clear where this was going Put aside your (definitely valid) opinions on gun control, we can't fix that problem if we allow these fascists to destroy our democracy first
Explicit Contrastive Framing	The pair uses overt contrast markers or alternative constructions—such as “but,” “instead,” “while,” “rather than,” or “not X, but Y”—to set competing claims, actions, or groups against each other.	1. I'd rather vote for another Trump than a corporate shmuck Democrat 2. Why is climate policy a success for special interests, but not the climate? 3. Ahhh. It's just like when poor people are addicts, they're deplorable criminals. But when rich people are addicts, they're going through a tough time. 4. Imagine if we had a party full of AOCs making progress, instead of a bunch of geriatrics threatening to stop confirming all of Trump's nominees sometime in the future maybe soon idk

Name	Definition	Examples
Extended Explanatory Argumentation	Posts develop a claim through multiple sentences or paragraphs that provide explanation, reasons, supporting details, examples, or justification rather than relying solely on a brief slogan or assertion.	1. ... We know the voters wanted a centrist to win because that's what happened. It's not different than Bernie endorsing Biden in the general election. Did you want him to run as an independent and split the vote against Trump? ... 2. No, it's worse than doing nothing. It makes the Democratic party look incompetent. It gives republicans even more reason to rally behind trump. It fires up trump's base like almost nothing else can. It also devalues actual real impeachment attempts. 3. No, it's an opportunity. Global warming, which will melt the ice, will create more arable land, expecially in Antarctica, this land can be used for to cultivate crops or cattle, and create new lands to colonise and exploit for natural resources. 4. I think America is great and always has been, I love America. I fought in a war to defend our way of life, I have raised successful children here and pulled my own self out of poverty ...
Hostile Outrage Venting	Posts primarily express anger or outrage through hostile denunciation, contempt, ridicule, insults, or profanity toward a target.	1. Fuck taxes. They literally took 500 dollars out my fucking check. 2. Im more pissed about all the rich assholes who got ppp loans forgiven why can't they pay their debts like every one else!?!?!?!? 3. That Jet should fucking stay in Qatar, the message it send about America is totally disgusting. Proud of being American and seeing this happening? I doubt. 4. Fuck that 100% of parents with disabled children are shitting bricks about their childrens life saving support thats disappearing
Limited Profanity Within Argument	The pair uses occasional mild profanity, slang, or insults within broader substantive commentary rather than sustained, standalone, or slur-heavy abuse.	1. ... I still believe in gun rights (with both background checks and mental evaluation) , but everything else I lean pretty fucking far left now lol. ... 2. ... much of the increase in their emissions occurred in order to build up their cleantech industries. I just wish that they hadn't built so many damn coal power plants over the last decade is all. 3. It is so damn difficult to get two particular things accomplished in the USA, climate action and gun safety legislation. ... 4. The government needs to leave scientists alone. Let them figure this shit out, and stop dicking around with universities. ...

Name	Definition	Examples
Opponent Claims Framed for Rebuttal	The post quotes, paraphrases, or rhetorically challenges an opposing claim, slogan, or premise and then argues against or undermines it.	1. A lady in her 60s told me last night that abortion access does not matter for anyone who has hit menopause. I disagree. Just because I'm too old to get pregnant does not negate the need to protect abortion access for younger women. ... 2. It's voter suppression because most mail voters are Democrat? That is a nonsequitur. You need to demonstrate that the new rule would prevent people from voting. 3. One of the greatest myths ever sold was that fossil fuels and agriculture would destroy the world. In reality, they feed families, power economies, and make modern life possible. 4. ... you are making the argument that if we do it that will give permission to Democrats to do it. That isn't how it works, they will do whatever they can get away with regardless of whether we give them precedent.
Partisan Collective Blame	Attributes broad social, political, economic, or violent harms to an entire political party, ideology, supporter base, or other large partisan out-group rather than to specific individuals or actions.	1. The Republican Party owns the upcoming recession. 2. Republicans are also the problem. They have blocked any kind of gun reform for the past 40 years of mass shootings. 3. Unfortunately this country must hit rock bottom then maybe just maybe the government will work to protect Americans but I seriously doubt it. We will become a third world country thanks to the republicans. 4. Or you can just look at the last 80 years of Republican presidents. They always leave higher unemployment job losses and recessions. It is what Republicans are and what they do. This will be no different.
Partisan Mockery and Taunting	Posts use sarcasm, ridicule, caricature, mock quotations, or taunting rhetorical questions to demean, humiliate, or belittle political opponents.	1. The irony of Trump sleeping every day is just too much. Sleepy Joe? 2. That's woke Jesus. We believe in the non-Biblical, definitely white, <i>American</i> Jesus who died on the cross for these stars and stripes. 3. Trump supporters used 8646, so they mean the same thing. So tell us conservatives, what DOES it mean again? 4. Any of this sound familiar? "Global warming is killing people," folks. Drownings have never happened before, anywhere around the world.
Personal Attacks on Political Figures	Posts directly target identifiable politicians, public officials, or associated supporters with hostile personal insults, character attacks, or derogatory labels.	1. Trump isn't a man, Trump is an infantile shit stain. 2. trump, the clueless, spineless, brainless, dickless asshole who somehow lied his way into the job as president of our country. He's a damned disgusting disgrace. ... 3. Surrender Schumer better not cave to help pass that shit this time. 4. JD Vance is a horrible person but he's also not popular with GOP folks. He's being tolerated because he's an accessory to Trump.

Name	Definition	Examples
Policy Advocacy and Tradeoff Arguments	Posts advocate, oppose, or critique specific government policies or reforms, often explaining their expected consequences, tradeoffs, implementation, or priorities.	1. Term limits don't discriminate between the good & the bad. They remove institutional memory & make legislators more dependent on lobbyists. I oppose term limits. 2. The Green New Deal for Housing matters because it links housing, labor, climate, and public ownership. That's the right instinct: solve social problems collectively, not by begging the market to be nicer. 3. If Europe were to invest in public transportations instead of enforcing stricter regulations on Engines , Carbon emissions would drop 4. I am generally in favor of trust busting to break up large corporations so they're not too big to fail. It's cheaper and more responsible.
Policy Consequence Warnings and Civic Mobilization	Posts express concern about governmental, institutional, or policy harms by citing consequences for rights, safety, fairness, or material well-being and often urge accountability, reform, voting, or other political action.	1. Vote blue for women's rights to stay secured. 2. Registering voters at the gun store! It is imperative that gun owners vote for candidates who will protect their second amendment! 3. Today is a perfect time for you to check your voter registration. This close to the election, you should check weekly. Vote blue to undo. 4. The United States of America is broken and neither Republicans nor Schumer and Jeffries are going to fix it. Voters must act wisely and bravely to begin to repair Republican damage.
Political Actor and Institution Criticism	Posts criticize political parties, leaders, ideological groups, or institutions for their policies, conduct, competence, corruption, or governing choices.	1. Newsom and Cuomo are a big part of what is wrong with the Democratic party. 2. It was inflation especially the price of houses. Many people are libertarian but they want government to fix problems. Joe Biden and the democrats did not appear to understand the problem. 3. Trump spent the last year gutting FEMA and delaying disaster funds. Americans are getting less protection from bigger storms—and facing an affordability crisis, leaving them less likely to be able to prepare and recover from disasters. 4. That second part “prevent it from happening again”, is what the Democrats did nothing about, when they had a majority, under Biden. The Democrats need new leadership as well. Schumer & Jefferies aren't up to the task.
Political and Institutional Targets	The pair primarily targets political parties, elected officials, ideological factions, government bodies, or related elite institutions and organized interests.	1. Thank your republican, both Trump and congress. FDT. 2. That Trump Foreign policy, really showing us how it's done. 3. The DNC hand selecting the candidates is what keeps getting Trump elected. 4. @BJP4India @narendramodi Inflation is breaking household budgets. People need solutions, not promises.

Name	Definition	Examples
Punitive Blanket Condemnation of Political Opponents	Posts broadly demonize political opponents or their supporter groups and advocate or wish for their punishment, exclusion, removal, or other punitive treatment.	1. To make democracy work people who show signs of fascism should be excluded from government jobs and voting. 2. Political party is not a protected class. We can refuse service. 3. I would like to see masked ICE agents banned as well 4. Get ‘em where it hurts. They obviously have enough money to pay a two digit fine, so instead take away their pride, bragging right, mode of transportation, and deadly assault weapon.
Quoted Slogans and Emphatic Framing	Posts use quoted or slogan-like political language, sometimes with capitalization, scare quotes, italics, or exclamation marks, to frame, attribute, critique, or comment on claims.	1. Leftists decided in 2024 that “saving Gaza” was more important than stopping Trump, so here we are. 2. It’s because republicans have the “fake news” ideology to hide behind. It is the solution to cognitive dissonance. 3. “Defund the Police” was not pragmatic; it cost the Dems dozens of Statehouses in 2020, which resulted in even worse gerrymandering. 4. Kinda like where’s our “regulated militia” NRA when our Federal Government starts kidnapping residents? The “don’t tread on me” folks are MIA.
Reasoned Political Policy Persuasion	Posts seek to persuade through substantive political or policy arguments, using explanations, justifications, evidence, examples, or rebuttals rather than merely asserting a position.	1. 45,000 Americans die from gun violence every year. 90% of us want universal background checks. States with stronger laws see 40% fewer deaths. We have the data. We have public support. What we lack is political courage to act on evidence. 2. Taxes on businesses actually encourage reinvestment into the company, since the taxes are on profits, so if you spend it, you dont pay the taxes. In the past high corporate taxes pushed up salaries & investment in research & development, aka innovation. 3. A PHEV with 50km electric range is still 90% petrol over its lifetime. Why give it the same incentives as a zero-tailpipe BEV? This isn’t smart policy its subsidizing fossil fuels in disguise. 4. We brought down violent crime rates significantly in the 90s primarily by incarcerating huge numbers of violent offenders. If someone commits a violent crime they should be imprisoned. . . .
Shouting-Style Emphatic Formatting	Posts use conspicuous all-caps, repeated exclamation or other emphatic punctuation, slogans, fragments, repetition, or intensifiers to convey a loud, urgent, confrontational tone.	1. Now for the love of God, democratic socialists of voting age: FUCKING VOTE. VOTE. VOTE FOR FUCKS SAKE. GOD FUCKING DAMNIT. <i>VOTE!</i> 2. DEMOCRATS ARE LITERALLY INSANE. THEY ALL NEED TO BE LOCKED AWAY IN AN INSANE ASYLUM. STOP THIS NONSENSE. 3. CAN THE USELESS SHITBAG WOMEN THAT RUN THE USA HR DEI THATS RACIST DEPARTMENTS EVERYWHERE RESIGN ALREADY NO. WHY?? 4. FUCK the New York Times sane washing bullshit. Fuck fuck fuck fuck fuck. . . .

Name	Definition	Examples
Specific Political and Institutional References	Posts use concrete names of politicians, parties, agencies, institutions, policies, legal concepts, or issue-specific political terminology rather than only broad political language.	1. Steve Hilton supports the second amendment. #California. 2. Yes. I'd vote Rubio over a lot of current dems. 3. If Trump wants to bomb Iran, Vance would be all for it! 4. @MichaelSteele I was done with Cornyn when he supported red flag laws.
Substantive Policy and Institutional Claims	Posts make concrete factual, causal, or interpretive claims about laws, government actions, institutions, political actors, or their social, economic, and rights-related consequences.	1. ... The date in which the new Congress takes effective is constitutionally set as the 3rd of Jan (though we deviate from it for convenience when everyone agrees). On that day, a new Speaker is elected. <i>That</i> Speaker swears them in. ... 2. Trump is already doing it with disaster aid. Something like 83% of requests made by GOP governors for aid are fulfilled and 23% for democratic governors. 3. It was, maybe not in name, but in fact. Every single policy goal of the Green New Deal was implemented by Biden over the course of his presidency with things like his infrastructure bill. 4. The only instances of Democrat voter fraud that I'm aware of in the last decade are the felons who tried to vote in Florida, after voters reinstated voting rights for felons, but before the state changed the actual law.
Sweeping Unsubstantiated Political Accusations	Posts make broad, categorical allegations that political opponents or leaders are corrupt, criminal, authoritarian, immoral, dishonest, or otherwise malign without qualifying evidence or nuance.	1. Trump is probably the most corrupt person in American history 2. Trump lies to keep his fanatical base on his side. 3. Trump is a liar, a cheater, and a fraud as the 'leader' of his party. First 2 his whole adult life, the last only relatively recently. 4. Pathetic attempt at whitewashing a legacy of corruption by McConnell

Results

We have evidence that suggests that people moderate on the civility of conversation and less on the substance of the conversation.

We find that the probability of a post being removed is correlated to increases in hostile language and negatively linked to material policy arguments. Substance-related references and mentions of political actors remain relatively constant.

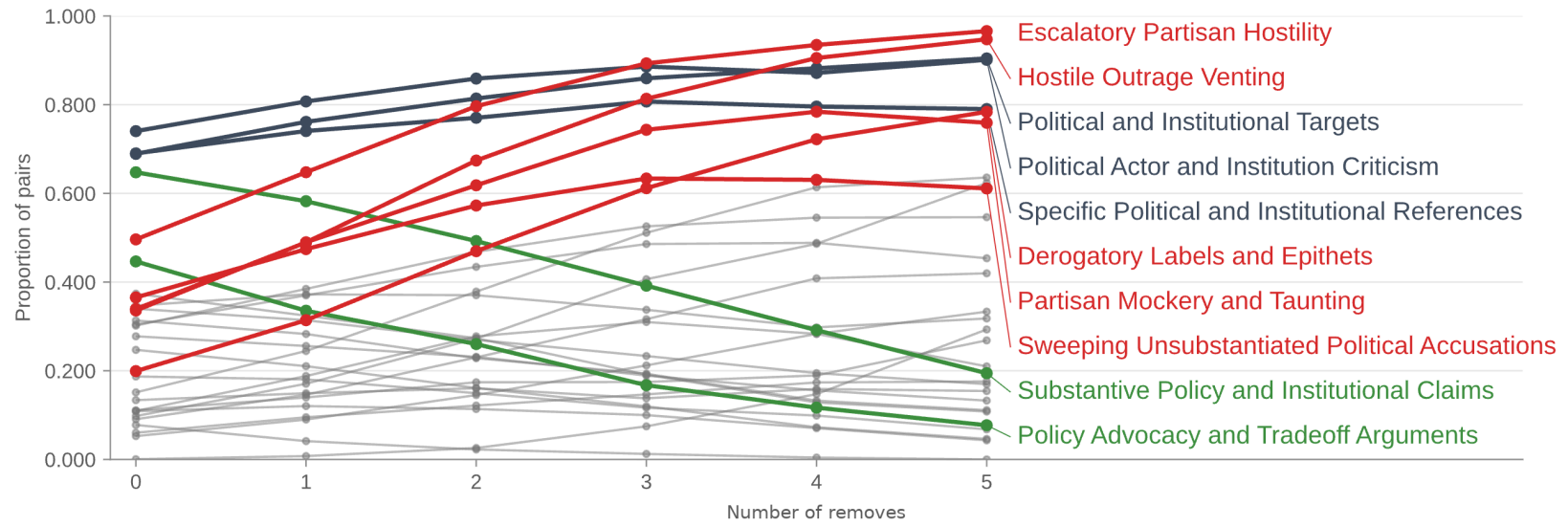


Figure 8: Topic Distribution by Keep/Remove

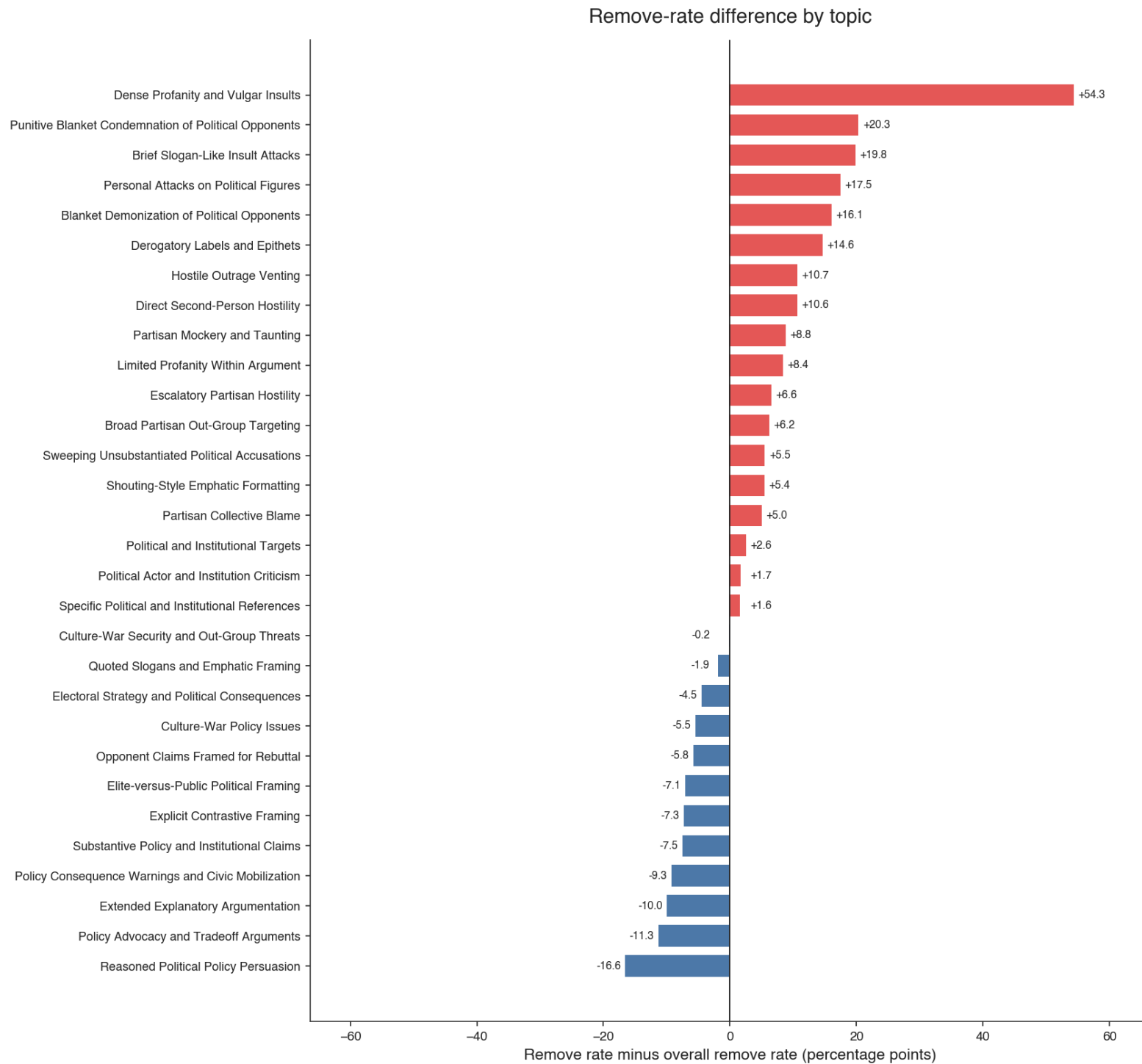


Figure 9: Remove-rate difference by topic

Each bar is that topic's remove rate minus the overall remove rate.

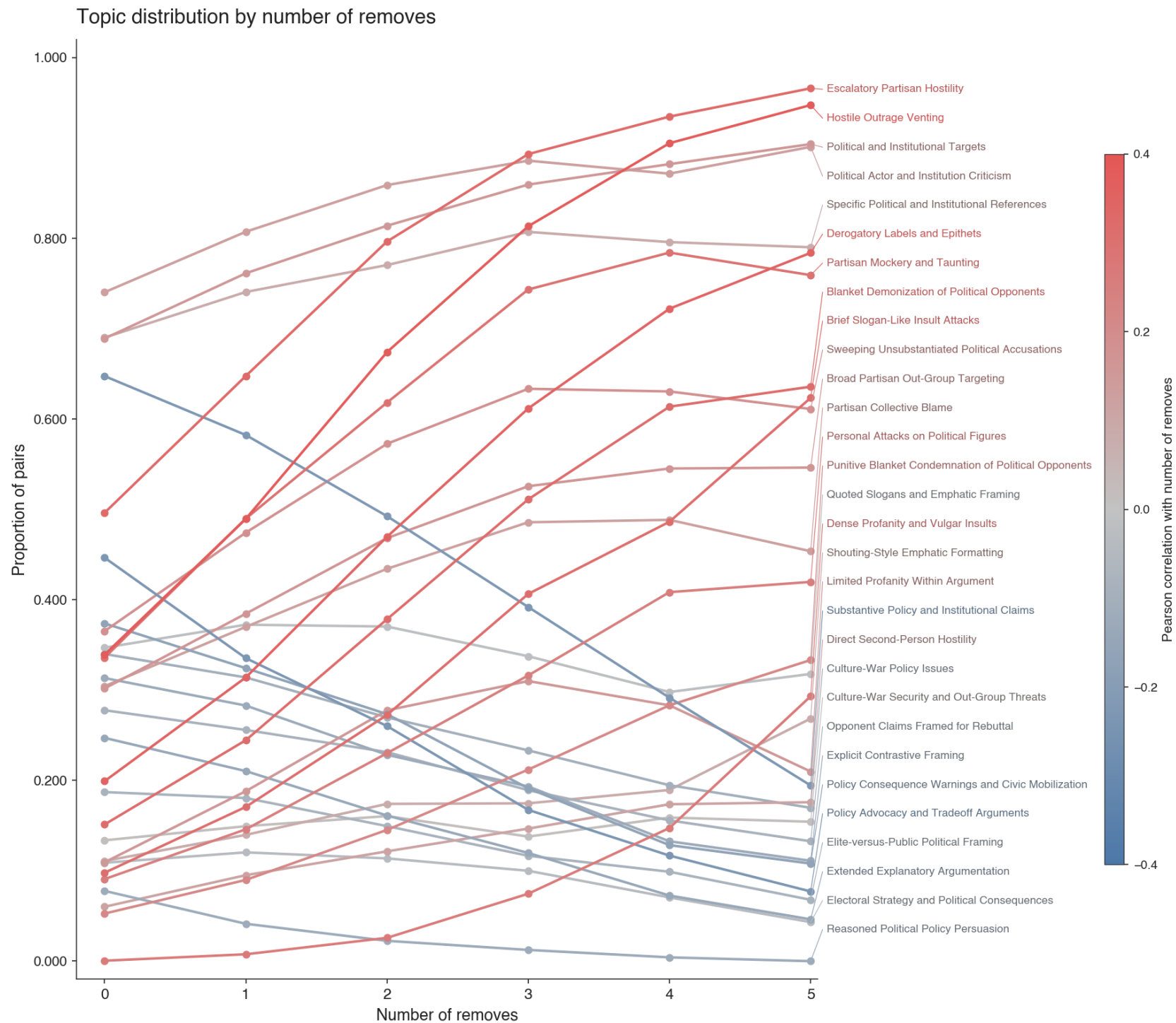


Figure 10: Topic distribution by number of removes

Line color is the Pearson correlation between whether the post has that topic and the number of remove votes.

Similarly, we find that as toxicity of the post increases, hostile language increases and policy argument decreases, while substance-related references and mentions of political actors remain relatively constant.

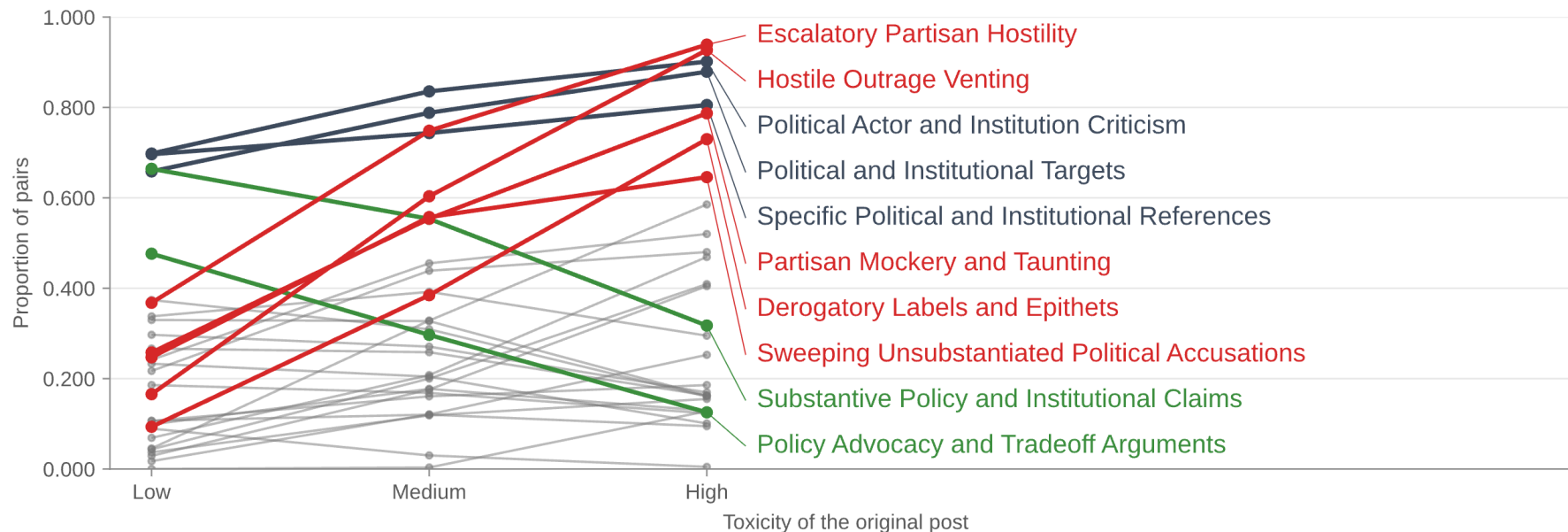


Figure 11: Topic Distribution by Toxicity

Training a binary classifier

Work is still ongoing and pending results

We train a variety of binary classifiers to predict the keep/remove task. We experimented with the following approaches:

1. Zero-shot LLM inference
2. Few-shot LLM inference
3. Few-shot prompt-tuned LLM inference
4. Fine-tuning an open-source LLM

We test across the following models:

- Amazon Nova
- Qwen 3 32B
- OpenAI GPT-5.6 Terra
- Claude Sonnet 5.5

For fine-tuning, we use Qwen3.8-27B. We deploy using vLLM and we use a quantized deployment.

Dataset

We evaluate our models across 3 versions of our dataset. We filter for posts with exactly 5 labels, to avoid ties in labels (n=13,992):

1. Unanimous posts (n=4,051): posts that were unanimously kept or unanimously removed across labelers.
2. Split posts (n=9,941): posts that were neither unanimously kept nor unanimously removed across labelers.
3. All posts (n=13,992): the total dataset of posts. For these posts, we take the majority label.

Prompts

Zero-shot prompt We use the same prompt that was shown to study participants, randomly shuffling the order of the original and mirror post:

We are developing a new social media platform designed specifically for political discussion. As part of this process, we'd like to understand how

The pairs are political mirrors of each other. This means that the mirror text recreates the original message from the opposite political stance.

Original Text:

I'm a bleeding-heart liberal, and I think the issue of abortion is obviously about protecting women's rights!

Mirror Text:

I'm a staunch conservative, and abortion is fully about the sanctity of human life before birth!

Notice that the mirror text changes the core message to match that of the opposite political stance. In other words, the mirror text is not a resp

Your job is to decide whether both posts in the pair should be allowed or removed from the platform.

When making your decisions, consider generally whether a post contributes to a healthy environment for political discussion, or whether it would b

Post 1: {post_1_text}

Post 2: {post_2_text}

Allow Or Remove?

Few-shot prompt We use a similar prompt as the zero-shot version, but we append 5 few-shot examples of post pairs that were kept and 5 few-shot examples of post pairs that were removed. We sampled posts that were either unanimously kept or unanimously removed, and model training and evaluations exclude these few-shot samples.

We are developing a new social media platform designed specifically for political discussion. As part of this process, we'd like to understand how

The pairs are political mirrors of each other. This means that the mirror text recreates the original message from the opposite political stance.

Original Text:

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Mirror Text:

I'm a staunch conservative, and abortion is fully about the sanctity of human life before birth!

Notice that the mirror text changes the core message to match that of the opposite political stance. In other words, the mirror text is not a resp

Your job is to decide whether both posts in the pair should be allowed or removed from the platform.

When making your decisions, consider generally whether a post contributes to a healthy environment for political discussion, or whether it would b

Post 1: {post_1_text}

Post 2: {post_2_text}

Here are examples of pairs of posts that human annotators remove:

1. Post 1: Conservative Christians are on the side of the capitalist class or billionaires and should be physically fought. #communism #socialism
2. Post 1: Fuck this smug DNC-groomed hypocrite. The day he actually fights for the Second Amendment will be the first day, since he's been kissin
3. Post 1: Woke, DEI, social justice, critical race theory, politically correct, civil rights, restorative justice, reparations... Just use the ha
4. Post 1: And why there are far fewer soy-brained libtard town hall meetings to 'splain why Biden's corruption and cognitive decline is good for
5. Post 1: Democrats will never be an honest decent party with integrity, generosity, and warmth - it's all about grasping for power and quite a b

Here are examples of pairs of posts that human annotators keep:

1. Post 1: If everyone who says something to the effect of "gun rights are important for hunters" understood how stupid the average Wisconsinite h
2. Post 1: For decades the US left told us we shouldn't impose American values on other countries - sovereignty matters, they said. But now they're falling all over themselves to copy Europe's socialist failures: crushing taxes, open
3. Post 1: always under siege Second Amendment. Lloyd Smucker has my Complete and Total Endorsement for Re-Election. Election Day is Tuesday, May
4. Post 1: The Secretary of State should be charged with voter suppression by making it harder for working people to cast their ballots in person.
5. Post 1: I take exception to some of how the definition of "assault weapon" is formed. New York, where I grew up, made significant regulations b

Allow Or Remove?

Optimized prompt We performed automated prompt tuning using DSPy and the GEPA optimization metric.

DSPy is a framework for writing a language-model program as typed inputs and outputs, then optimizing that program. Here the program was one predictor. Its inputs

were the two post texts. Its outputs were a Boolean remove decision and a remove probability. GEPA is a reflective prompt optimizer. On each iteration it draws a small batch, runs the current program, and reads a written explanation of what the program got wrong. A reflection call then proposes a new instruction. GEPA keeps that proposal only when it strictly improves the batch score. For the DSPy and GEPA optimization, we use GPT 5.6-Terra.

We started with our few-shot prompt and iterated. We fixed the few-shot examples and edited the system prompt. The optimized system prompt is as follows:

You are reviewing a pair of political mirror posts for a political social media platform. The posts express analogous messages from opposing viewpoints.

Decide whether the pair supports healthy political discussion.

Allow substantive political speech, including criticism of politicians, parties, policies, institutions, and movements; claims that policies cause harm.

Remove if either post contains:

- Targeted hostile abuse, harassment, threats, calls for harm, dehumanization, or degrading/contemptuous language.
- Sweeping attacks on political groups or their members.
- Obscene, aggressively dismissive, taunting, mocking, or inflammatory language directed at opponents or their views.
- Unsupported assertions or insinuations that a named person committed serious criminal, sexual, or similarly grave misconduct. A bare accusation is not enough.

Distinguish criticism from abuse:

- Allow criticism focused on actions, policies, consequences, public statements, or documented conduct, when not abusively framed.
- Remove language whose main effect is to insult, humiliate, provoke hostility toward, or contemptuously dismiss a political person, group, or viewpoint.
- Do not remove solely for strong political disagreement. Remove only when abusive, threatening, degrading, inflammatory, or gravely accusatory as defined above.

Make one binary decision for the pair:

- Allow only if both posts should remain.
- Remove if either post should be removed.

{Few-shot examples}

Return only the decision.

Allow Or Remove?

Baselines

Zero-shot baselines Across all posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.426029	0.550529	0.786388	0.292152
Qwen 3 32B	0.498773	0.649800	0.821429	0.358108
OpenAI GPT-5.6 Terra	0.316304	0.800100	0.217992	0.576135
Claude Sonnet 5.5	0.223602	0.803459	0.133423	0.689895
Jev	0.529322	0.712621	0.761792	0.405561

Across unanimous posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.256932	0.603061	0.902597	0.149784
Qwen 3 32B	0.395946	0.779314	0.951299	0.250000
OpenAI GPT-5.6 Terra	0.478528	0.937053	0.379870	0.646409
Claude Sonnet 5.5	0.382134	0.938534	0.250000	0.810526
Jev	0.493617	0.853123	0.941558	0.334487

Across split posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.467645	0.529122	0.772932	0.335236
Qwen 3 32B	0.517117	0.597022	0.806391	0.380589
OpenAI GPT-5.6 Terra	0.294281	0.744291	0.199248	0.562633
Claude Sonnet 5.5	0.203249	0.748416	0.119925	0.665971
Jev	0.535016	0.655367	0.740977	0.418649

Zero-shot baselines — across all posts

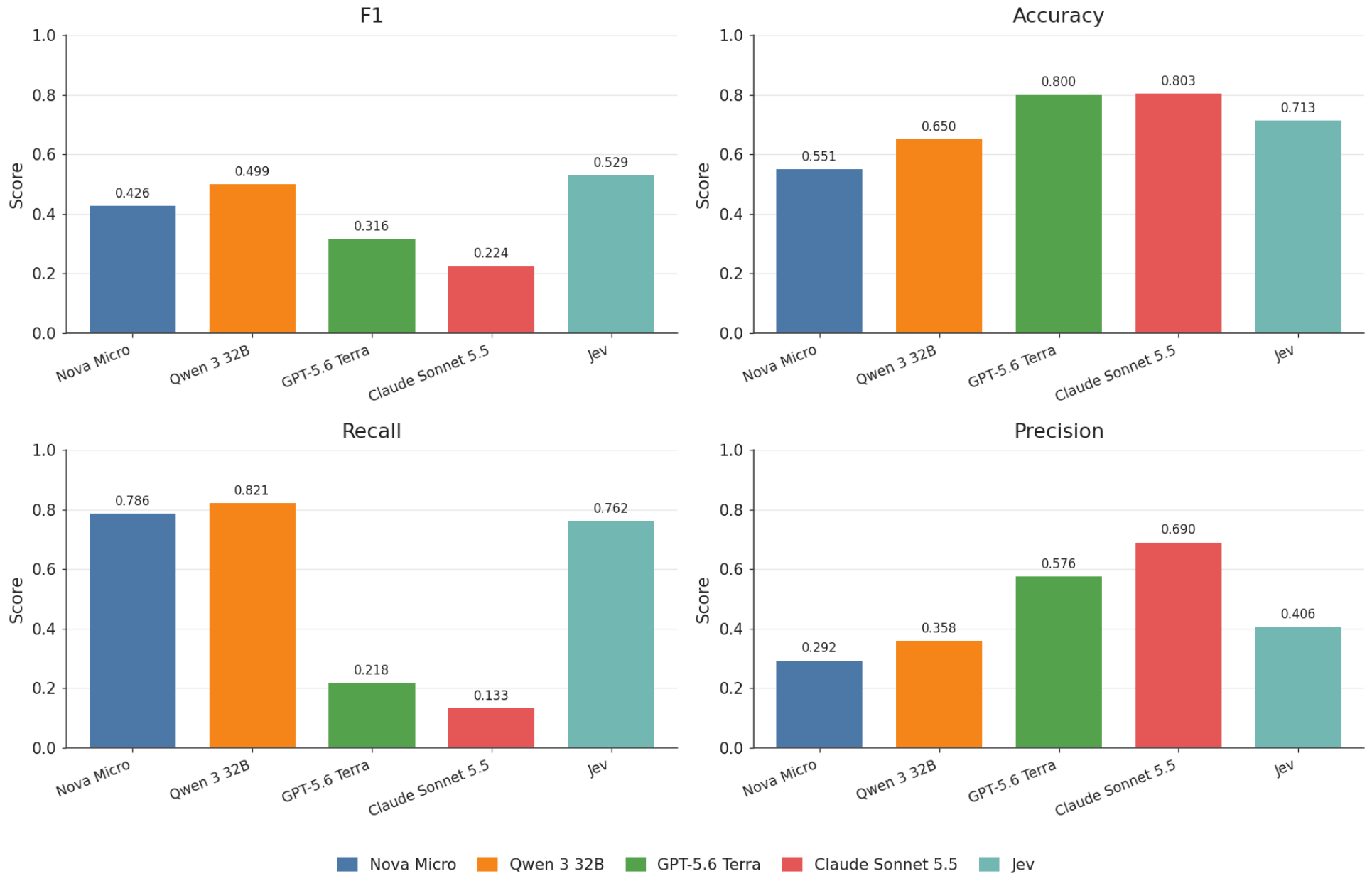


Figure 12: Zero-shot baselines across all posts

F1, accuracy, recall, and precision for the zero-shot baselines across all posts. Bar color identifies the model.

Few-shot baselines Across all posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.428947	0.580253	0.743425	0.301435
Qwen 3 32B	0.463378	0.566812	0.881996	0.314234
OpenAI GPT-5.6 Terra	0.338110	0.805176	0.234659	0.604692
Claude Sonnet 5.5	0.330366	0.806392	0.225219	0.619666
Jev	0.561910	0.787517	0.642616	0.499214

Across unanimous posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.274428	0.654968	0.862745	0.163164
Qwen 3 32B	0.305225	0.668067	0.964052	0.181315
OpenAI GPT-5.6 Terra	0.526316	0.942165	0.424837	0.691489
Claude Sonnet 5.5	0.511931	0.944390	0.385621	0.761290
Jev	0.664251	0.931290	0.898693	0.526820

Across split posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.464521	0.549844	0.729699	0.340706
Qwen 3 32B	0.496046	0.525601	0.872556	0.346521
OpenAI GPT-5.6 Terra	0.312448	0.749422	0.212782	0.587747
Claude Sonnet 5.5	0.307005	0.750226	0.206767	0.595883
Jev	0.547683	0.729001	0.613158	0.494842

Few-shot baselines — across all posts

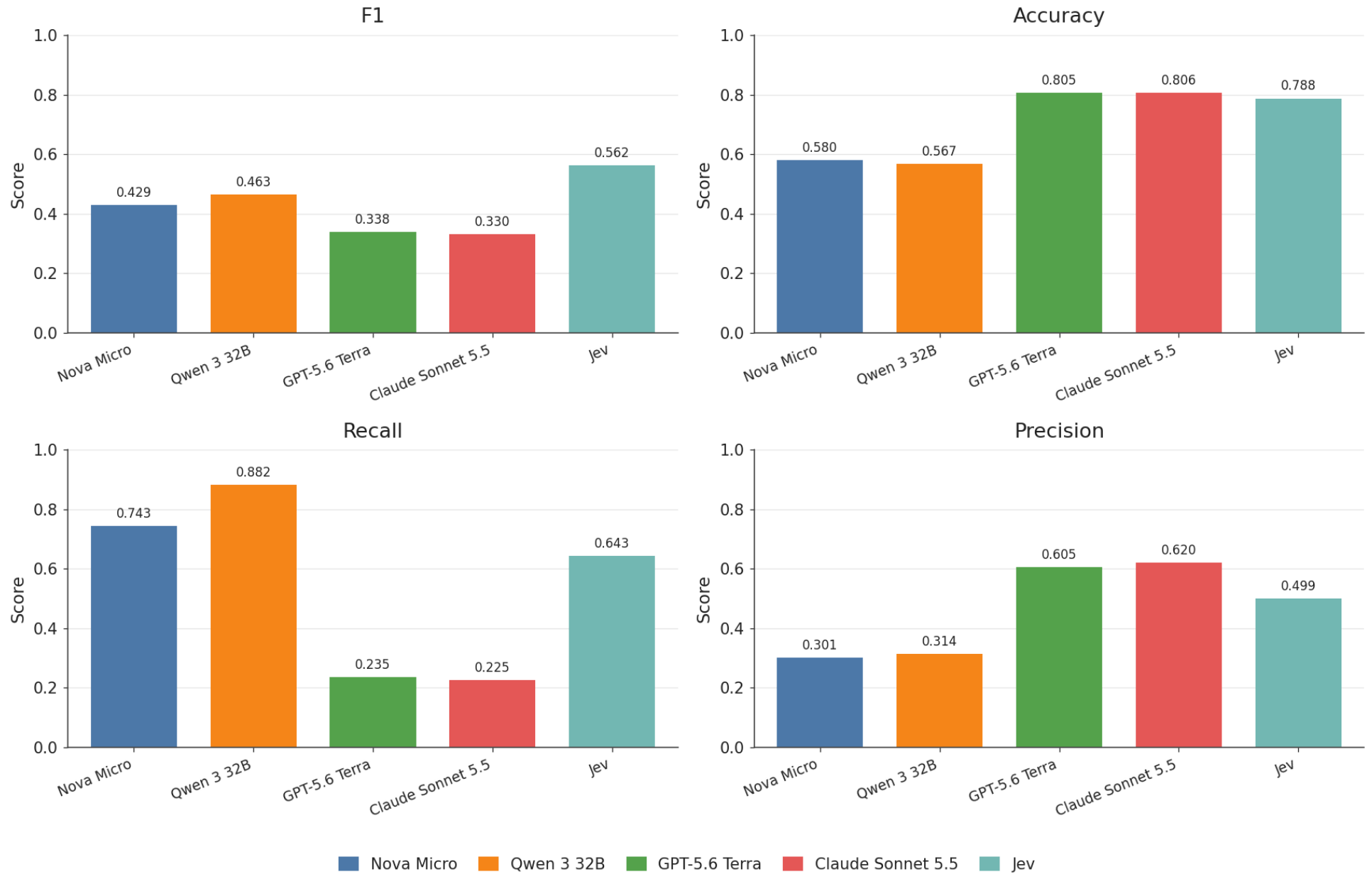


Figure 13: Few-shot baselines across all posts

F1, accuracy, recall, and precision for the few-shot baselines across all posts. Bar color identifies the model.

Automated prompt tuning

We used DSPy and GEPA to optimize the original few-shot prompt. We used GPT 5.6-Terra to review and rewrite the prompts. We trained and tested the optimization on only unanimous keep/remove labels. The new few-shot prompt improved F1 from 0.68 to 0.84 on the hold-out set.

Model Version	F1	Accuracy	Recall	Precision
Original	0.680851	0.754098	0.533333	0.941176
Selected	0.835821	0.819672	0.933333	0.756757

We rerun Jev, Amazon Nova Micro, and Qwen 3 32B on this optimized prompt (GPT-5.6 Terra and Claude Sonnet 5.5 underperform these models and are 10x the price).

Across all posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.281617	0.803031	0.182063	0.621404
Qwen 3 32B	0.466164	0.640738	0.739717	0.340313
Jev	0.542148	0.739043	0.728591	0.431682

Across unanimous posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.452489	0.940188	0.326797	0.735294
Qwen 3 32B	0.355202	0.759516	0.875817	0.222776
Jev	0.549451	0.888532	0.898693	0.395683

Across split posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.259358	0.747209	0.165414	0.600273
Qwen 3 32B	0.487348	0.592395	0.724060	0.367277
Jev	0.541099	0.678201	0.709023	0.437486

Prompt tuning vs few-shot — across all posts

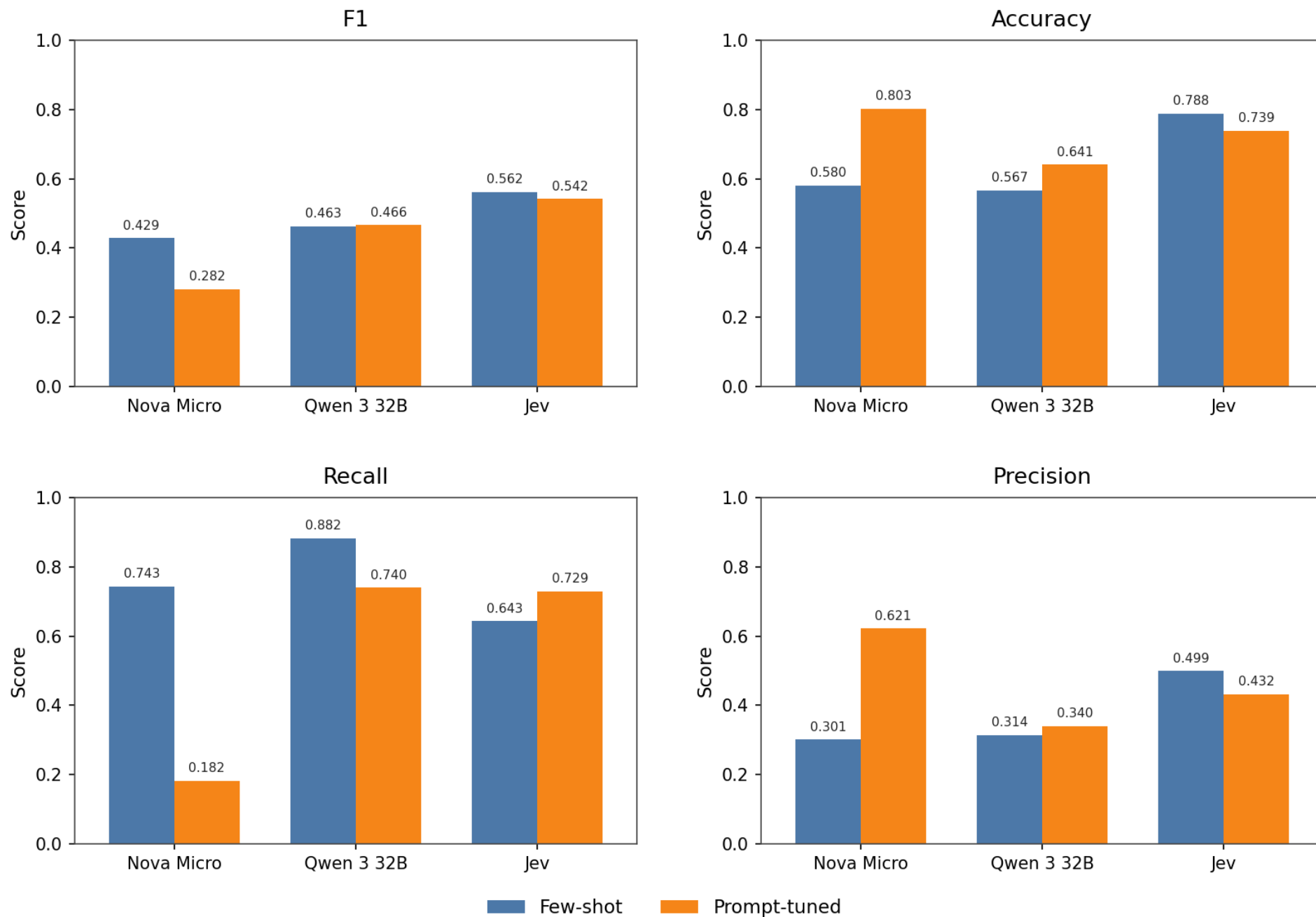


Figure 14: Prompt tuning vs few-shot — across all posts

F1, accuracy, recall, and precision across all posts, comparing few-shot and prompt-tuned scores for Amazon Nova Micro, Qwen 3 32B, and Jev.

Fine-tuning a classifier

We fine-tuned **Qwen3.5-4B** on the keep/remove task. Training used **Qwen3.5-4B** with LoRA rank 128 and alpha 32, for one epoch at learning rate 2e-4. The sequence limit was 4,096 tokens, in bf16. The per-device batch size was 8. The optimizer was AdamW with a linear schedule, no warmup, weight decay 0, and gradient clipping at 1.0. Gradient checkpointing was on. LoRA dropout was 0. We trained the model on Hugging Face and AWS Sagemaker, using the open-source Hugging Face version of **Qwen3.5-4B**. Training ran on one A100 GPU, using Flash Attention. Inference ran on Sagemaker **m1.g5.xlarge** using Flash Attention, bf16, prefix caching, and vLLM.

We developed three versions of fine-tuned models:

1. **Ablation 1:** A model fine-tuned on only the unanimous labels (n=4,051)
2. **Ablation 2:** A model fine-tuned on only the split labels (n=9,941)
3. **Ablation 3:** A model fine-tuned on both the unanimous and split labels (n=13,992)

We evaluated how each model performed as well on the those same three slices of our dataset:

1. **Dataset 1:** Unanimous labels (n=4,051).
2. **Dataset 2:** Split labels (n=9,941).
3. **Dataset 3:** Both the unanimous and split labels (n=13,992)

Unanimous adapter

Dataset	Total Samples	Accuracy	Precision	Recall	F1
Unanimous	4051	0.9975	0.9686	1.0000	0.9840
Split	9941	0.7790	0.6483	0.3805	0.4795
All labels	13992	0.8423	0.7025	0.4447	0.5447

Split adapter

Dataset	Total Samples	Accuracy	Precision	Recall	F1
Unanimous	4051	0.9657	0.7666	0.7890	0.7776
Split	9941	0.9439	0.8762	0.9203	0.8977
All labels	13992	0.9502	0.8650	0.9067	0.8853

All-label adapter

Dataset	Total Samples	Accuracy	Precision	Recall	F1
Unanimous	4051	0.9938	0.9521	0.9675	0.9597
Split	9941	0.9506	0.8880	0.9331	0.9100
All labels	13992	0.9631	0.8945	0.9367	0.9151

Summary of key modeling results

Each bar is the best model in that stage by F1 on the all-labels evaluation, the full set of posts with five labels (n=13,992). The four models are Jev in zero-shot (F1 0.529322), Jev in few-shot (F1 0.561910), Jev after few-shot prompt tuning (F1 0.542148), and the Qwen3.5-4B all-label adapter after fine-tuning (F1 0.9151). Accuracy, recall, and precision are that same model’s scores, so a stage can lead on F1 while trailing on recall or precision. In the baseline tables this slice is “Across all posts”; in the fine-tuning table it is “All labels.”

Best model by stage

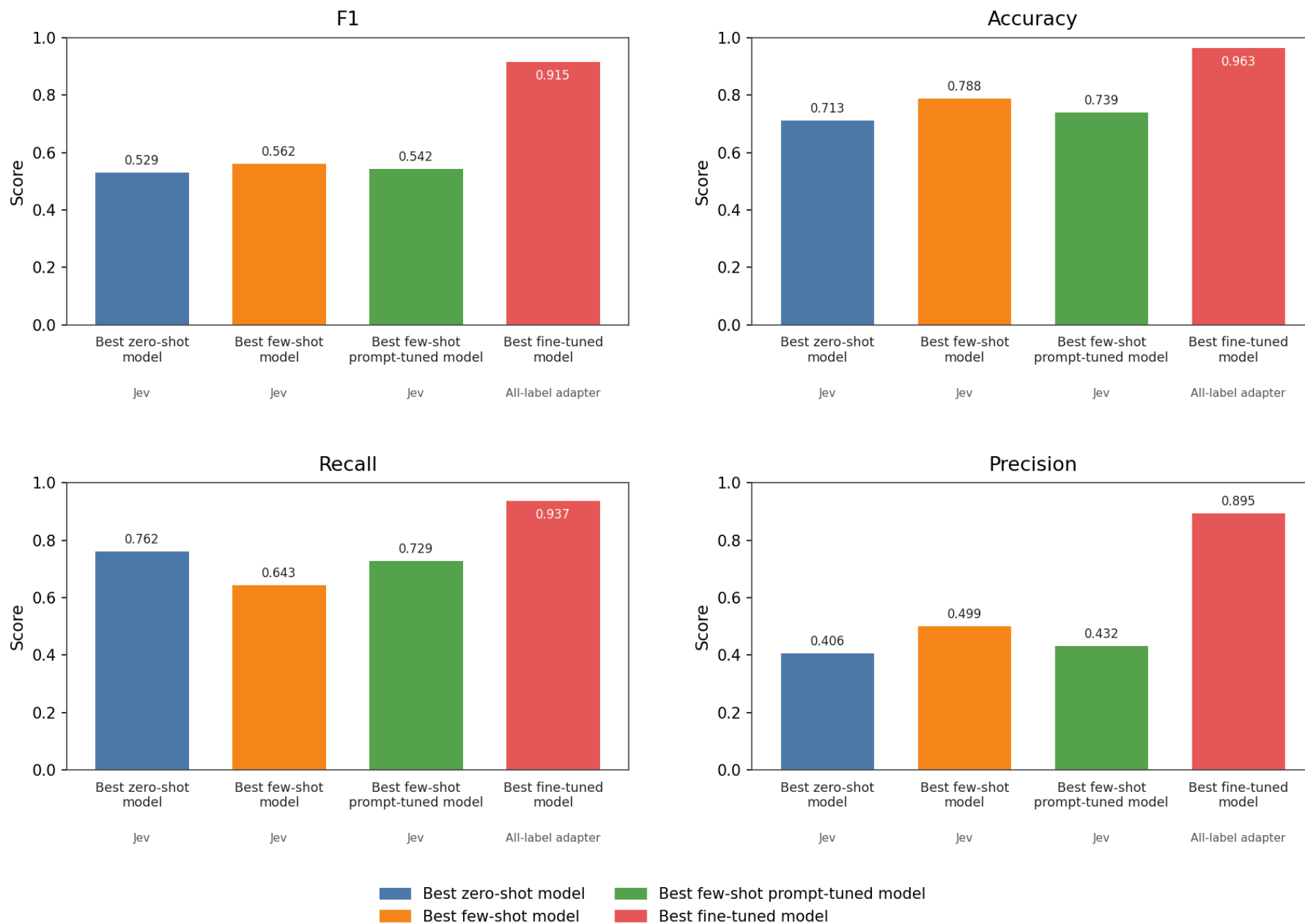


Figure 15: Best model by stage

F1, accuracy, recall, and precision for the best model in each stage on the all-labels evaluation.

(Ongoing) Training a calibrated classifier

Work is still ongoing and pending results

One of the shortcomings of building a binary classifier is that it imposes a discrete label on an inherently uncertain task. In a non-trivial amount of tasks, people themselves were uncertain of what the label should have been. We turn this into the following prediction task:

$p(\text{label}) = p(\text{proportion of 5 raters that would, on average, remove this post})$

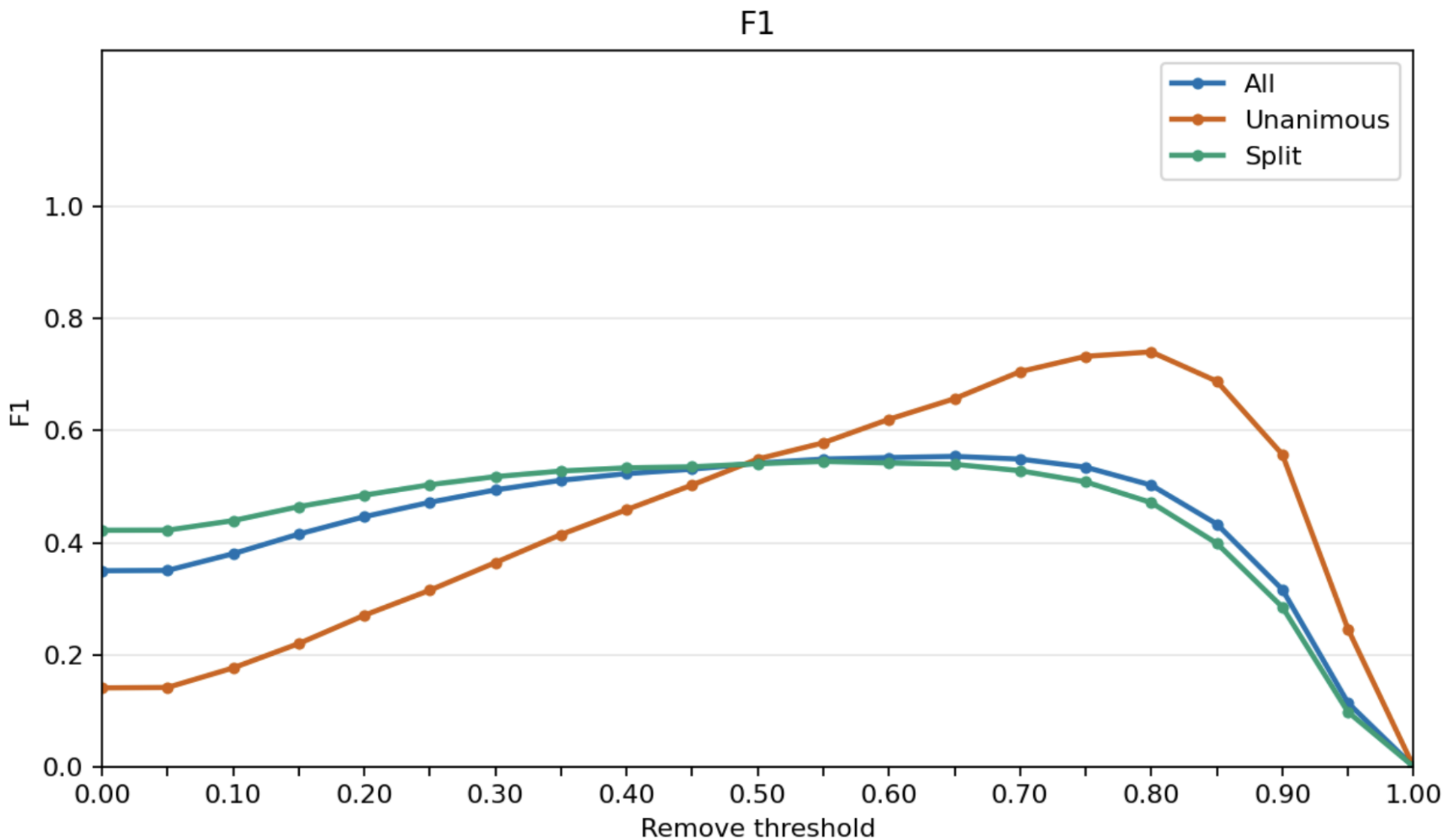
We present several approaches for this:

1. **Few-shot prompt-tuned LLMs that return their own probabilities:** We ask LLMs to report their own probabilities.
2. **Few-shot prompt-tuned calibrated classifier (Jev):** We prompt-tune Jev using DSPy and GEPA.
3. **An ensemble approach.** (Add more details)
4. **A calibrated classifier.** We can develop a classifier that returns a probability, and we can then set an arbitrary threshold to generate keep/remove decisions. This is inspired by how the Perspective API was built and follows past work on reinforcement learning with calibration rewards (RLCR).

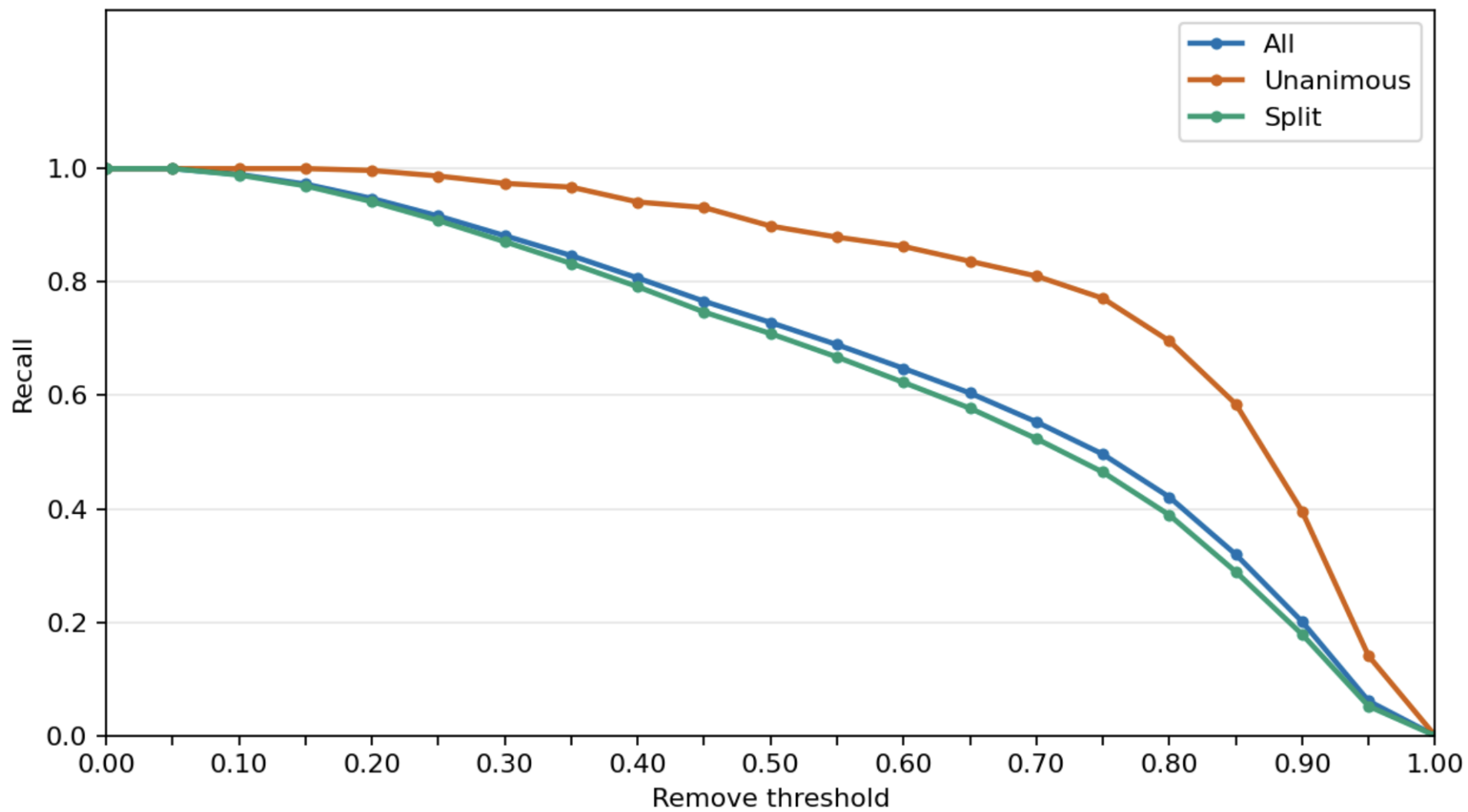
Prompt-tuning a calibrated classifier

One of the shortcomings of building a binary classifier is that it imposes a discrete label on an inherently uncertain task. In a non-trivial amount of tasks, people themselves were uncertain of what the label should have been. We use the optimized prompt generated by DSPy and GEPA and re-run the Jev classifier.

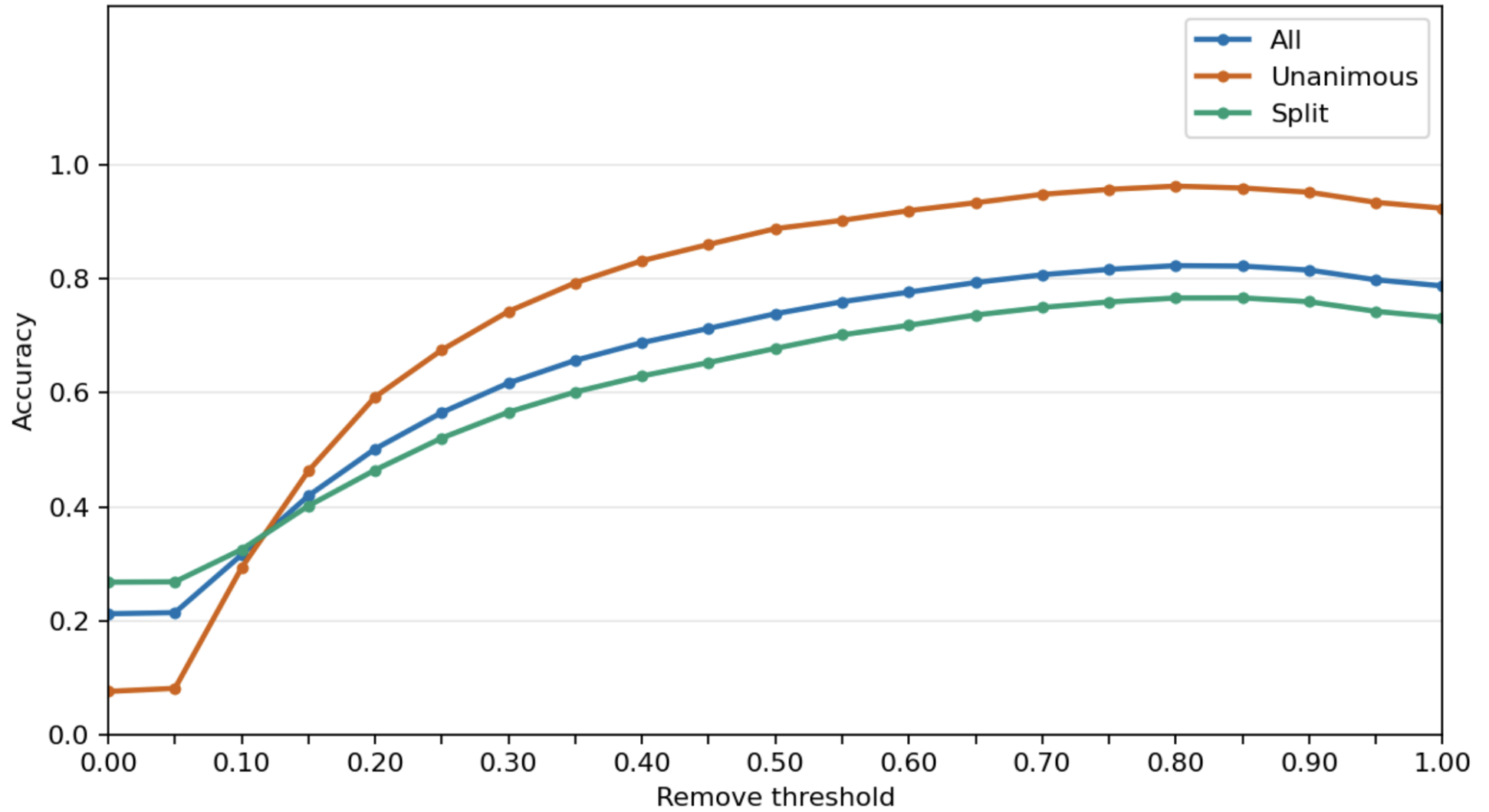
We see that the classifier’s performance on our task is sensitive to the threshold used for classification. Jev, unlike traditional LLM-based classifiers, can return calibrated probabilities (LLMs are notoriously poor at calibrating their own confidence).



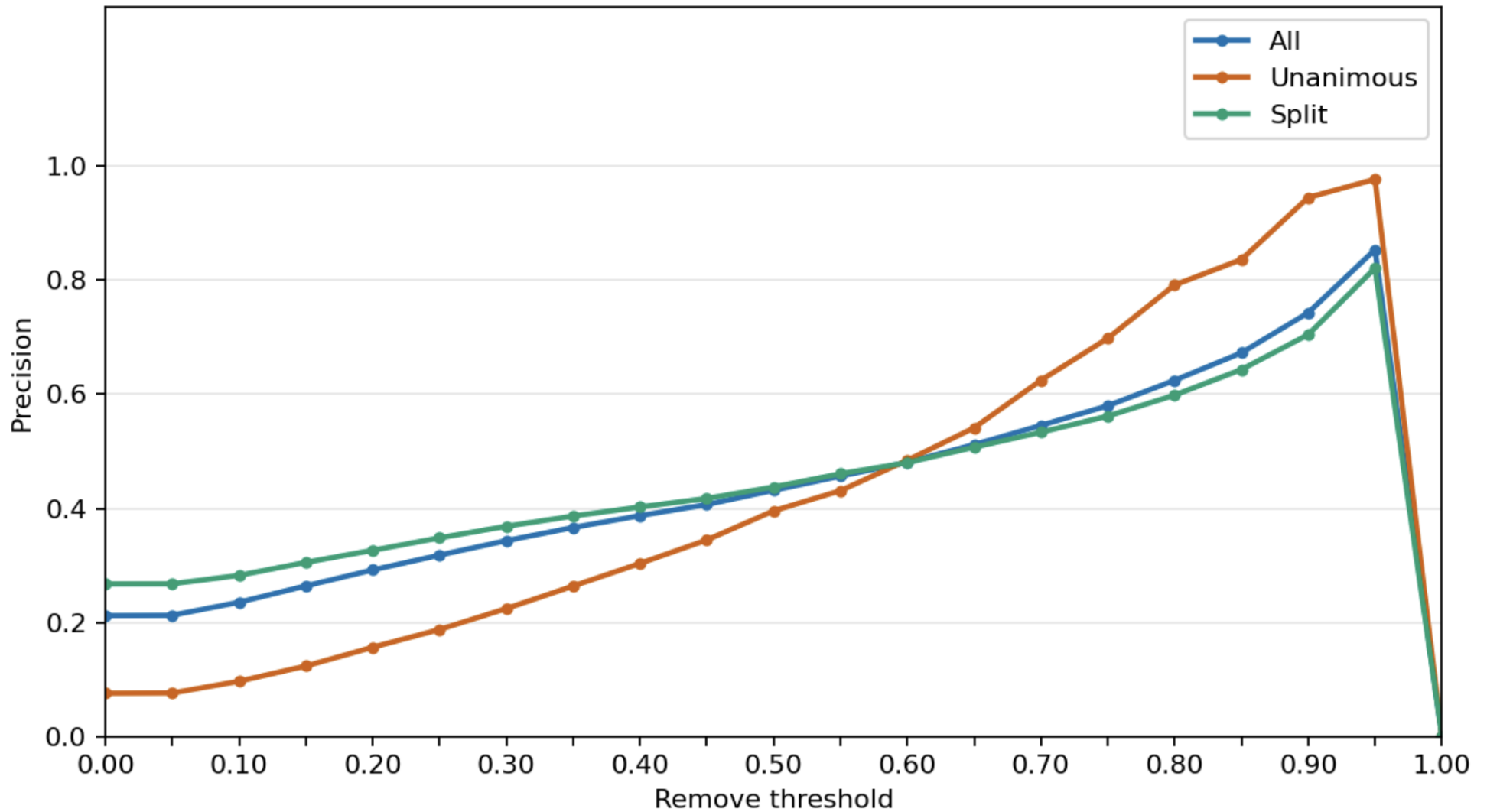
Recall



Accuracy



Precision



Fine-tuning a calibrated classifier

Work is still ongoing and pending results

Calibrated classifier fine-tuning results

Work is still ongoing and pending results

Which high-toxicity posts are kept

High toxicity posts are kept at 49.8% across 4,983 posts. That number is the average of each post's own keep rate. The stored majority label counts a tie as remove. With that label, 2,311 posts are labeled keep (46.4%) and 2,672 posts are labeled remove (53.6%).

These counts use the stored topic assignments and the stored feature labels. Topic names below are the short names from the tables above.

Among topics with at least 30 high-toxicity posts, the keep rate runs from 31.1% to 59.5%. Anti Trump harassment is kept at 31.1% (140 posts, 26 majority keep). Criticism of Republicans is kept at 40.5% (228 posts). MAGA and institutions is kept at 44.1% (314 posts). US and Iran is kept at 59.5% (45 posts), and climate and fossil fuels is kept at 58.8% (47 posts). Trump media and lying is the largest of these topics, with 403 posts, and it is kept at 52.9% (207 majority keep). Border and immigration is kept at 53.6% (137 posts), the same high-toxicity cell as in the chart above.

2,091 high-toxicity posts are ungrouped, which is 42.0% of this set. Their keep rate is 50.8%. They are 43.6% of the majority keep posts and 40.6% of the majority remove posts. Another 674 posts sit in grouped topics with fewer than 30 high-toxicity posts, and their keep rate is 51.2%. The chart leaves those two rows off, because each row is about the same share of both groups. Each bar is that topic's share of the majority keep posts or of the majority remove posts.

Topic	Posts	Majority keep	Keep rate
Anti Trump harassment	140	26	31.1%
Criticism of Republicans	228	75	40.5%
Anti fascism messaging	86	31	42.0%
MAGA and institutions	314	112	44.1%
Abolish ICE	95	38	44.2%
Trump media and lying	403	207	52.9%
Climate and fossil fuels	47	29	58.8%
US and Iran	45	32	59.5%
Ungrouped	2,091	1,007	50.8%

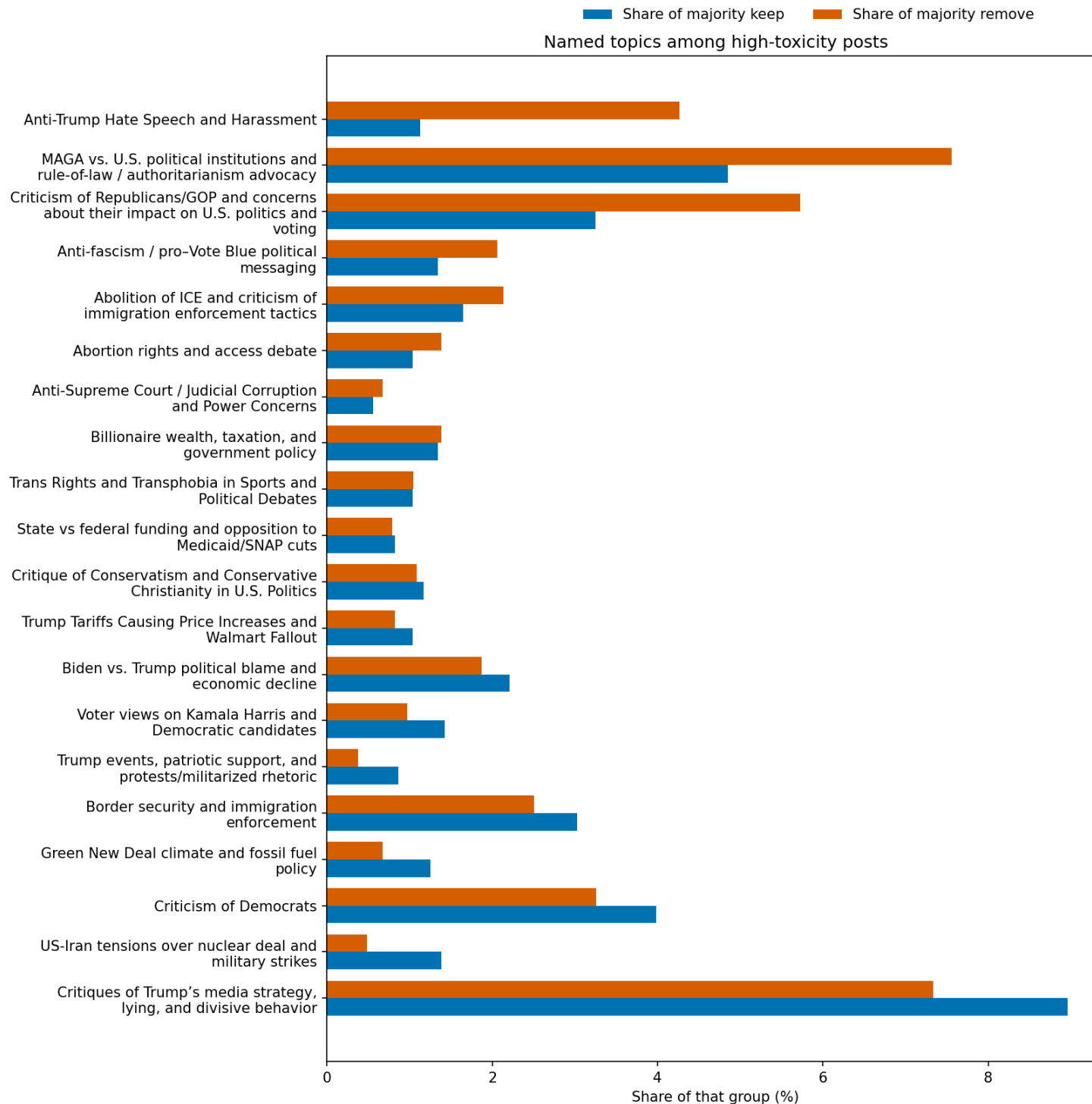


Figure 16: Named topics among high-toxicity posts

The feature labels are the same present-or-absent labels as the feature charts above. Limited Profanity Within Argument is on 50.4% of the majority keep posts and 33.0% of the majority remove posts. The keep rate is 55.8% when that feature is present and 45.6% when it is absent. Substantive Policy and Institutional Claims are on 39.0% of the majority keep posts and 25.6% of the majority remove posts. The keep rate is 56.2% when that feature is present and 46.8% when it is absent.

Brief Slogan-Like Insult Attacks are on 38.1% of the majority keep posts and 54.6% of the majority remove posts. The keep rate is 44.6% when that feature is present and 54.4% when it is absent. Dense Profanity and Vulgar Insults are on 5.4% of the majority keep posts and 19.2% of the majority remove posts. The keep rate is 32.3% when that feature is present and 52.3% when it is absent.

Escalatory Partisan Hostility is on 91.6% of the majority keep posts and 96.3% of the majority remove posts. Hostile Outrage Venting is on 89.7% of the majority keep posts and 95.6% of the majority remove posts. The posts without Escalatory Partisan Hostility are kept at 61.2%. The posts without Hostile Outrage Venting are kept at 62.3%.

Feature	Share of kept	Share of removed	Keep rate if present	Keep rate if absent
Limited Profanity Within Argument	50.4%	33.0%	55.8%	45.6%
Substantive Policy and Institutional Claims	39.0%	25.6%	56.2%	46.8%
Brief Slogan-Like Insult Attacks	38.1%	54.6%	44.6%	54.4%
Dense Profanity and Vulgar Insults	5.4%	19.2%	32.3%	52.3%



Figure 17: Feature gap inside high-toxicity posts